FUSIONCORE v_0

Phase 3 — Physics-Aware Feature Engineering

Notebook 03 | Technical Deep-Dive Analysis

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1. Abstract

This report presents a rigorous technical analysis of the physics-aware feature engineering pipeline implemented in FusionCore v_0, Phase 3 (Notebook 03). The pipeline operates on the unified, regime-normalised dataset FD00u produced by Phase 2, comprising 129,331 training rows and 31,028 validation rows across 709 distinct turbofan engine trajectories drawn from the NASA C-MAPSS FD001–FD004 corpus (Saxena & Goebel, 2008) **[1]**. Two principal engineering mandates are addressed: the systematic resolution of thirteen outstanding gaps carried forward from Phase 1 and Phase 2 technical reports, and the construction of a 91-feature physics-informed dynamic feature vector for downstream RUL estimation.

Phase 3 operates two distinct verification frameworks which must not be conflated. The first is the Gap Resolution Audit (Part A), which addresses the thirteen outstanding engineering gaps inherited from Phases 1 and 2. Seven gaps carry MANDATORY status and are each closed via inline programmatic assertions embedded within their respective execution cells:

- G1 (composite key uniqueness),
- G2 (temporal ordering),
- G5 (regime_id one-hot encoding),
- G6 (s7 cross-fault-mode scale differential),
- G7 (mechanically linked sensor pairs s8/s13 and s9/s14),
- G8 (extreme z-score excursion disposition), and
- G10 (multivariate health index).

The six remaining gaps carry DIAGNOSTIC status and are documented with quantitative diagnostics and formally deferred:

- G3 (dataset class imbalance) to Phase 4,
- G4 (regime row imbalance) to Phase 4,
- G9 (s16 partial-dead status) retained with ablation deferred,
- G11 (regime out-of-distribution guard) to FusionCore v_1 deployment,
- G12 (survival analysis framework) — Kaplan-Meier and Cox Proportional Hazards diagnostic completed; integration into scheduling decision layer deferred to Phase 5, and

- G13 (per-fault-mode-family UWL recalibration) — family-specific P-F interval distributions quantified and adaptive thresholds calibrated; operational deployment deferred to Phase 5.

The second framework is the Feature Engineering Gate Verification (Part B), which independently confirms the integrity of the output feature matrix via six programmatic acceptance gates applied after all engineering steps are complete:

- Gate 1 confirms active sensor count at 17, consistent with Phase 2 regime-dead classification (s1, s5, s18, s19 confirmed dead),
- Gate 2 confirms total feature count at 91, exceeding the formula minimum of 81,
- Gate 3 verifies feature matrix dimensional consistency across train $(129{,}331 \times 91)$ and validation $(31{,}028 \times 91)$ splits
- Gate 4 confirms zero NaN values across both splits
- Gate 5 confirms all three virtual sensor distributions are non-degenerate and
- Gate 6 verifies all cumulative fatigue indices are strictly non-decreasing in accordance with Miner's Rule (Miner, 1945) [2], with no monotonicity violations detected within a tolerance of -1×10^{-10}.

All seven gap resolution assertions and all six feature engineering gates pass on the executed run. All thirteen gaps are addressed—seven resolved, six documented with justified deferrals. The 91-feature matrix is constructed via a four-step pipeline:

i. Core Manifold retaining all 24 base features (3 operational settings + 21 sensors)
ii. Kinematic Expansion applying first-order differences, rolling means, and rolling standard deviations to the 17 active sensors, yielding 51 additional features
iii. Three Thermodynamic Virtual Sensors encoding (a) compressor pressure-temperature ratio, (b) thermal efficiency proxy, and (c) EGT margin drift and
iv. Three Cumulative Fatigue Indices implementing a Miner's Rule damage accumulation proxy for EGT, core speed, and HPC pressure.

The pipeline is designed for compatibility with the Phase 4 XGBoost baseline and SOTA benchmark architectures (DeepAR, TFT, PatchTST). Six primary residual concerns are identified and formally transferred to Phase 4 for resolution.

2. Scope and Objectives

Phase 3 of the FusionCore v_0 pipeline serves a dual mandate. The first mandate is the systematic resolution and documentation of every outstanding gap identified in the Phase 1 Exploratory Data Analysis and the Phase 2 Zero-Leakage Normalisation reports. The second mandate is the construction of the complete physics-aware dynamic feature vector that constitutes the input to the Phase 4 XGBoost baseline model and all subsequent SOTA benchmark comparators.

The notebook operates exclusively on the unified, regime-normalised dataset FD00u produced by Phase 2, comprising four C-MAPSS subsets (FD001–FD004) combined into a single train and validation partition. All engineering operations preserve the zero-leakage principle established in Phase 2: no statistical parameter is computed using validation-set observations. The gap resolution component (Part A) and the feature construction component (Part B) are executed in strict sequence, with a six-gate verification block confirming pipeline integrity before outputs are persisted.

3. Dataset Provenance and Partition

The FD00u dataset loaded from Phase 2 parquet outputs contains 30 columns encompassing unit identifiers, cycle index, three operational settings, 21 raw sensor channels, the piece-wise linear RUL target (capped at 125 cycles) (Heimes, 2008) [3], the Phase 2 provenance subset label, the integer regime identifier, and the global subset-origin tag. The partition dimensions confirmed at notebook load are presented in Table 1.

Table 1 — FD00u Partition Dimensions at Phase 3 Intake

Partition	Row Count	Engine Count	Columns	Subset Distribution
Internal Train	129,331	567	30	FD001: 16,561 (12.8%) FD002: 43,464 (33.6%) FD003: 20,012 (15.5%) FD004: 49,294 (38.1%)
Validation	31,028	142	30	Held out from Phase 2 stratified split: mirror subset proportions

The composite key (`subset_origin`, `unit_id`) is the unique engine identifier throughout Phase 3, resolving the fundamental ambiguity that `unit_id` integers are reused across the four C-MAPSS subsets (G1). All groupby operations apply this composite key to prevent cross-engine data contamination during temporal feature construction.

4. Part A — Gap Resolution Analysis

Thirteen gaps were carried forward from Phase 1 and Phase 2 technical reports. The classification of each gap by severity determined the resolution pathway: MANDATORY gaps required code-level resolution before the feature matrix could be constructed; DIAGNOSTIC gaps required quantitative documentation and a justified disposition. Table 2 provides the consolidated audit summary.

Table 2 — Phase 3 Gap Resolution Audit

Gap	Description	Severity	Disposition
G1	`unit_id` global uniqueness	MANDATORY	RESOLVED — composite key `(subset_origin, unit_id)` enforced
G2	Row temporal ordering	MANDATORY	RESOLVED — sort by `(subset_origin, unit_id, cycle)` with monotonicity assertion
G3	Dataset class imbalance	DIAGNOSTIC	DOCUMENTED — deferred to Phase 4 (XGBoost robust to imbalance; stratified sampling evaluated in Phase 4)
G4	Regime row imbalance	DIAGNOSTIC	DOCUMENTED — deferred to Phase 4 (inverse-frequency weighting for gradient-based models)
G5	`regime_id` ordinal encoding	MANDATORY	RESOLVED — one-hot encoded to `regime_0–regime_5`; integer column dropped
G6	s7 cross-fault-mode sigma differential (3.93x)	MANDATORY	RESOLVED — binary `fault_mode_family` indicator added; confirmed 1.00x ratio post-normalisation
G7	Mechanically linked pairs (s8/s13, s9/s14)	MANDATORY	RESOLVED — VIF quantified; ratio features constructed; raw sensors retained
G8	Extreme z-score excursions	MANDATORY	RESOLVED — cycle-position analysis confirms terminal-dominance; no Winsorisation applied
G9	s16 (farB) partial-dead	DIAGNOSTIC	DOCUMENTED — $IQR = 0$, active only in FD002/FD004; ablation in Phase 4

Gap	Description	Severity	Disposition
G10	Multivariate health index	MANDATORY	RESOLVED — composite index $mean(s4, s7, s9)$ constructed in z-space
G11	Regime assignment OOD guard	DIAGNOSTIC	DOCUMENTED — distance-to-centroid quantified; full guard deferred to v_1 deployment pipeline
G12	Survival analysis	DIAGNOSTIC	DOCUMENTED — Kaplan-Meier and Cox PH diagnostic completed; scheduling integration deferred to Phase 5
G13	Per-fault-mode-family UWL recalibration	DIAGNOSTIC	DOCUMENTED — family-specific P-F distributions quantified; adaptive thresholds calibrated; deployment deferred to Phase 5

4.1 G1 & G2 — Composite Key Enforcement and Temporal Ordering (Cell 7)

The C-MAPSS dataset assigns unit_id integers independently within each of the four subsets, meaning that `unit_id=1` in FD001 and `unit_id=1` in FD002 refer to two entirely distinct engine trajectories. Failure to account for this ambiguity would corrupt any groupby-based temporal operation (rolling windows, first-order differences, cumulative sums), producing cross-engine contamination that constitutes a form of data leakage.

Resolution was achieved by defining the composite key `UNIT_KEY = [subset_origin, unit_id]` and applying this key in all groupby calls throughout the notebook. A deterministic sort over `(subset_origin, unit_id, cycle)` was applied, and two assertions were programmatically verified: (i) zero duplicate `(subset_origin, unit_id, cycle)` tuples across the entire dataset, and (ii) strictly monotonically increasing cycle values within every engine trajectory. Both assertions passed for the 129,331-row training partition (567 engines) and the 31,028-row validation partition (142 engines).

4.2 G3 — Dataset Class Imbalance (Cell 8)

The row distribution across subsets is substantially non-uniform. FD004 contributes 38.1% of all training rows, whilst FD001 contributes only 12.8%. This disparity arises from the operational complexity of each subset: FD002 and FD004 are multi-regime datasets (six K-Means regimes) whilst FD001 and FD003 operate under a single fixed regime. The number of engines and the distribution of failure cycle lengths differ accordingly.

Table 3 — G3 Subset Row Distribution and Terminal Fraction (Internal Train)

Subset	Row Count	Proportion (%)	Fault Modes	Regime Count
FD001	16,561	12.8	Single (HPC degradation)	1
FD002	43,464	33.6	Single (HPC degradation)	6
FD003	20,012	15.5	Dual (HPC + Fan)	1
FD004	49,294	38.1	Dual (HPC + Fan)	6
Terminal $(RUL < 30)$	17,010	13.2	—	—

The terminal fraction of 13.2%[1] is computed at the row (cycle-observation) level; all engines in FD00u reach failure by construction. Each engine contributes one row per operating cycle, meaning a long-trajectory engine of 300 cycles contributes 270 observations to the healthy regime and only 30 to the terminal window, whilst a short-trajectory engine of 50 cycles contributes 20 healthy rows and 30 terminal rows. The consequence is that when the model processes any given observation during training, 86.8% of the time it is presented with healthy operating data, and only 13.2% of the time with a terminal degradation signal. The critical degradation behaviour — the very dynamics the model must learn most precisely — therefore appears relatively rarely in the row-level training corpus. This structural imbalance is analogous to the class imbalance problem in binary classification: the model must infer RUL dynamics from a sparse terminal signal embedded within abundant healthy-operating data, a condition that is not remedied by the run-to-failure construction of the dataset and which persists regardless of engine count.

Disposition: XGBoost uses gradient boosting, which iteratively re-weights residuals, providing inherent robustness to output-space imbalance [4]. Subset-stratified sampling will be evaluated empirically in Phase 4 if SHAP analysis or residual diagnostics reveal systematic bias against terminal predictions. No corrective action is taken at Phase 3.

Figure 1 — Row Count Distribution and RUL Density by Subset (Internal Train)

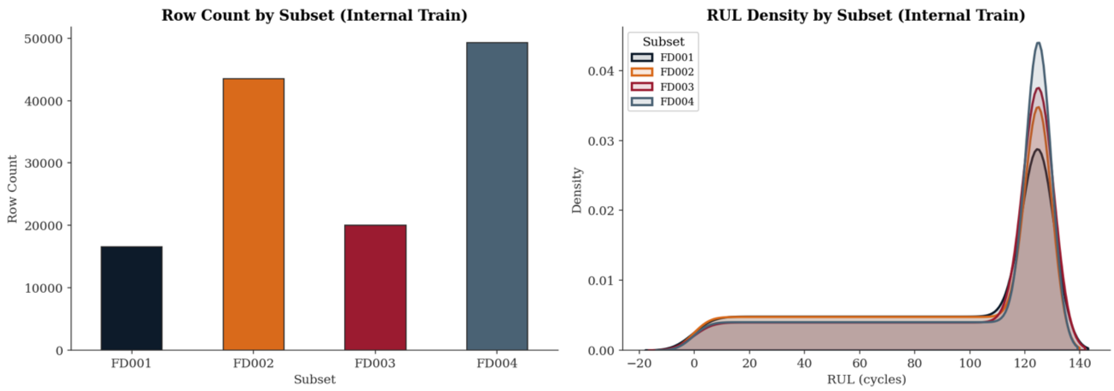


Figure 1: The left panel confirms the structural row imbalance across FD00u: FD004 dominates the corpus at approximately 49,000 rows, whilst FD001 contributes the fewest at approximately 16,500 rows — a disparity of roughly 3:1 driven by the greater number of engines and longer average trajectories in the multi-regime subsets . The right panel reveals why this imbalance carries modelling implications beyond simple sample size: all four subsets share a near-identical RUL density profile, with a pronounced mass concentrated at the piecewise-linear plateau ceiling of 125 cycles and a sharp left-side truncation at end of life[2]. The plateau arises directly from the clipped RUL target construction, where all cycles beyond 125 cycles from failure are assigned a capped value. The consequence is that the majority of row-level observations — irrespective of subset — are drawn from the plateau regime rather than the degradation window, reinforcing the terminal sparsity problem identified in Table 3. The near-perfect overlap of the four density curves further confirms that the RUL target distribution is structurally consistent across subsets, meaning the class imbalance is a property of the dataset construction, not a subset-specific artefact.

[1] The terminal threshold of RUL < 30 cycles is operationally calibrated to the minimum actionable maintenance lead time established in Phase 2 Gate 3: a median P-F interval of 36 cycles against a hard minimum of 30 cycles, reflecting realistic slot-booking and parts-provisioning constraints in commercial MRO operations (Nowlan & Heap, 1978) [20]. Rows satisfying this condition represent engines within the critical degradation window and constitute the primary signal of interest for downstream RUL estimation.

[2] The negative RUL values visible in the KDE tails to the left of zero are a smoothing artefact of the kernel density estimator and carry no physical meaning — no engine in the dataset has a negative remaining useful life.

4.3 G4 — Regime Row Imbalance (Cell 9)

Within the multi-regime subsets (FD002, FD004), the K-Means regime assignment produces a non-uniform distribution of rows across the six operational regimes. The overall distribution across all training data is shown in Table 4.

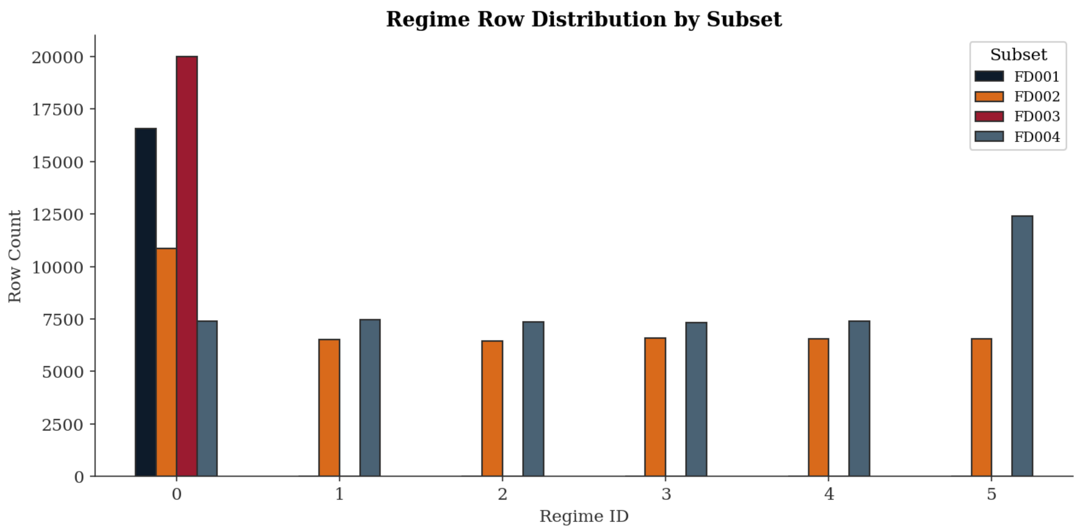
Table 4 — G4 Regime Row Distribution by Subset (Internal Train)

Regime	FD001	FD002	FD003	FD004	Total	Overall (%)
0	16,561	10,857	20,012	7,375	54,805	42.4
1	0	6,507	0	7,447	13,954	10.8
2	0	6,450	0	7,371	13,821	10.7
3	0	6,572	0	7,307	13,879	10.7
4	0	6,535	0	7,385	13,920	10.8
5	0	6,543	0	12,409	18,952	14.7

Regime 0 commands 42.4% of all training rows because FD001 and FD003 — single-regime subsets — are entirely assigned to Regime 0 by the Phase 2 K-Means pipeline, with zero contribution to any other regime. This is a structural feature of the C-MAPSS dataset arising from the fixed operational settings of the single-condition subsets and does not constitute a modelling error. Within the multi-regime subsets, FD002 distributes broadly uniformly across all six regimes. FD004 is similarly balanced for Regimes 0 through 4, but Regime 5 carries a disproportionate 12,409 rows — approximately $1.67\times$ the FD004 per-regime mean for Regimes 1–4 — the origin of which is not resolvable from the row distribution alone.

Disposition: Diagnostic. No corrective action is taken at Phase 3. Inverse-frequency regime weighting[3] is relevant for gradient-based and attention-based models that may underfit minority regimes and is formally deferred to Phase 4, where the SOTA benchmark architectures (DeepAR, TFT, PatchTST) may require regime-aware sampling strategies. XGBoost in Phase 4 does not require this intervention.

Figure 2 — Regime Row Distribution by Subset (Internal Train)



[3] Inverse-frequency regime weighting assigns each training observation a weight inversely proportional to its regime's row count, such that minority regimes contribute equally to the training signal relative to the dominant Regime 0. For a regime r containing n_r rows out of N total rows across R regimes, the weight is: $w_r = \frac{N/R}{n_r}$

Figure 2 reveals the structural origin of the Regime 0 dominance more clearly than the aggregate table alone. FD001 and FD003, being single-regime subsets, contribute their entire row counts exclusively to Regime 0 — 16,561 and 20,012 rows respectively — with zero representation in Regimes 1 through 5. FD002 distributes broadly uniformly across all six regimes at approximately 6,500 rows per regime. FD004 mirrors this balance for Regimes 0 through 4 at approximately 7,300–7,450 rows each but exhibits a pronounced asymmetry in Regime 5 at 12,409 rows — approximately $1.67\times$ the FD004 per-regime mean for Regimes 1–4. Whether this asymmetry reflects a genuinely higher prevalence of the Regime 5 flight condition in the dual-fault FD004 operating envelope, or is a K-Means boundary artefact, is not resolvable from the row distribution alone and is noted as an open diagnostic question.

4.4 G5 — regime_id Ordinal Encoding (Cell 10)

The integer regime_id column assigned in Phase 2 encodes an implied ordinal relationship $(0 < 1 < 2 < \ldots)$ that has no physical basis. K-Means cluster assignments are nominal: regime 3 is not "greater" than regime 2 in any meaningful engineering sense. Presenting a nominal variable as an integer to a tree-based or linear model may produce spurious splits or coefficient estimates that exploit the artificial ordering.

> Resolution: The integer `regime_id` was one-hot encoded using `pd.get_dummies`, producing six binary indicator columns (`regime_0` through `regime_5`) with dtype int. The original integer column was dropped. The one-hot representation is lossless and unambiguous for any downstream learner. The raw operational settings op1, op2, and op3 are simultaneously retained in the core manifold, providing the continuous regime signal independently of the discrete cluster assignment.

4.5 G6 — s7 Cross-Fault-Mode Sigma Differential (Cell 11)

s7 measures HPC outlet pressure (P30) and is one of the three primary degradation indicators in the FusionCore health monitoring framework alongside s4 (EGT) and s9 (Nc). Its behaviour across the four C-MAPSS subsets is not uniform, and the reason is structural: FD001 and FD002 are single-fault subsets in which only the HPC degrades, whilst FD003 and FD004 are dual-fault subsets in which both the HPC and the fan degrade simultaneously. The fan degradation mode introduces additional pressure variance into s7 readings that is physically unrelated to HPC health — it is variance from a second, concurrent fault source superimposed on the primary degradation signal.

Phase 2 quantified this differential at $3.93\times$ when measured as the ratio of within-regime standard deviations between dual-fault and single-fault populations prior to z-normalisation. If this scale differential were carried into the feature matrix uncorrected, the model would be implicitly learning different response magnitudes for what is nominally the same sensor, purely as a function of which fault modes happen to be active in a given subset. This constitutes a systematic structural confound rather than a genuine physical signal.

Per-regime, per-subset z-normalisation in Phase 2 resolves the scale problem directly by design. Following normalisation, both populations are standardised to unit variance within each regime. The sigma ratio measured at Phase 3 intake is $1.00\times$ (single-fault sigma: 1.0000, dual-fault sigma: 1.0000), confirming the normalisation step has fully equalised the within-regime scale of s7 across fault-mode families.

However, normalisation does not alter the underlying physics. In a single-fault engine, s7 movement in z-space reflects HPC degradation exclusively. In a dual-fault engine, the identical z-score is a mixture of HPC degradation and fan degradation — two signals with different shapes, different progression rates, and different relationships to RUL. After normalisation both populations produce numerically identical standard deviations, but the two numbers carry structurally different physical meaning. A model presented with only the normalised s7 value has no mechanism to distinguish between these two interpretive contexts and will conflate fundamentally different degradation dynamics into a single learned response function.

To preserve this structurally important distinction in the feature matrix, a binary indicator `fault_mode_family` is introduced: value 0 for FD001/FD002 (single-fault, HPC degradation only) and value 1 for FD003/FD004 (dual-fault, HPC and fan degradation). This flag provides the model — particularly a tree-based architecture such as XGBoost, which partitions the feature space via threshold splits — with the information necessary to learn separate response functions for s7 conditioned on fault mode family, without requiring two separate models or any restructuring of the feature matrix. The distribution across the internal training partition is: single-fault = 60,025 rows; dual-fault = 69,306 rows. The near-equal split is a favourable result, ensuring the model receives adequate exposure to both degradation regimes and is not biased towards either family by sampling imbalance.

Disposition: G6 resolved. The fault_mode_family binary indicator is added to the feature matrix and will be retained through Phase 4. Its importance relative to direct sensor features will be evaluated via SHAP analysis in Phase 4.

4.6 G7 — Mechanically Linked Sensor Pairs & VIF Analysis (Cell 12)

The physical relationship between corrected and uncorrected turbomachinery speed measurements is governed by the corrected speed equation from thermodynamic cycle theory (Walsh & Fletcher, 2004) [5]:

$$NR_x = \frac{N_x}{\sqrt{\theta}}, \qquad \theta = \frac{T_{\text{total}}}{T_{\text{ref}}}$$

where N_x is the physical shaft speed (rpm), NR_x is the corrected shaft speed, T_{total} is the total inlet temperature, and $T_{\text{ref}} = 288.15$ K is the ISA standard atmosphere reference temperature. For the C-MAPSS sensors, s8 = N_f (physical fan speed) and s13 = NR_f (corrected fan speed); s9 = N_c (physical core speed) and s14 = NR_c (corrected core speed). Within any single operating regime, T_{total} is held approximately constant by the K-Means regime assignment, rendering θ approximately constant and producing a near-perfect linear relationship between each physical and corrected speed pair. This constitutes structural multicollinearity — a condition imposed by the measurement physics rather than by statistical accident.

Multicollinearity was quantified using the Variance Inflation Factor (VIF), following the standard regression-based formulation of (Saxena & Goebel, 2008) [1]:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination obtained by regressing feature j on all remaining features. For the simplified two-variable case applicable here, this reduces to:

$$\text{VIF}_j = \frac{1}{1 - r^2}$$

where r is the Pearson correlation coefficient between the linked pair. Results are presented in Table 5.

Table 5 — G7 VIF Analysis for Mechanically Linked Sensor Pairs

Physical Sensor	Corrected Sensor	Physical Equation	Pearson r	VIF
s8 — Nf (Fan Speed, rpm)	s13 — NRf	$NR_f = N_f/\sqrt{\theta}$	0.9447	9.3
s9 — Nc (Core Speed, rpm)	s14 — NRc	$NR_c = N_c/\sqrt{\theta}$	0.9464	9.6

VIF values of 9.3 and 9.6 place both pairs in the moderate-to-severe multicollinearity category, approaching but not exceeding the conventional severity threshold of 10.0 (Montgomery et al., 2012) **[6]**. The values fall below this threshold rather than approaching infinity because per-regime z-normalisation in Phase 2 introduces small perturbations: $\sqrt{\theta}$ is approximately but not exactly constant within each regime, preserving a residual non-linear component in the relationship.

Whilst tree-based models such as XGBoost are robust to multicollinearity in coefficient estimation, VIF at this level depresses individual feature importance scores and introduces ambiguity into SHAP-based attribution in Phase 4 (Lundberg & Lee, 2017) **[7]**. The model has no principled basis for distributing importance between two near-identical features and may assign arbitrary proportions across training runs, rendering the importance scores unreliable as physical diagnostics.

Resolution: Ratio features are constructed to encode the correction factor $\sqrt{\theta}$ directly as an explicit, named feature:

$$\rho_f = \frac{s8}{s13} = \frac{N_f}{NR_f} \approx \sqrt{\theta_f}, \qquad \rho_c = \frac{s9}{s14} = \frac{N_c}{NR_c} \approx \sqrt{\theta_c}$$

Both raw sensor pairs are simultaneously retained in the feature matrix. The model therefore has access to the physical speed, the corrected speed, and the correction factor as three distinct, interpretable quantities[4].

Disposition: G7 resolved. Ratio features ρ_f and ρ_c added to the feature matrix. All four raw sensors retained. Phase 4 SHAP analysis mandated to audit ratio feature contributions against raw sensor importance.

Advisory — s9/s14 Ratio Instability: The ratio ρ_c exhibits an extreme standard deviation of 650.60 against a mean of 2.3604. Following per-regime z-normalisation, NR_c (s14) has a near-zero mean within each regime. When the denominator approaches zero the ratio becomes unbounded:

$$\rho_c = \frac{s9}{s14} \to \infty \quad \text{as} \quad s14 \to 0$$

A safe-division guard is applied:

$$\rho_c = \begin{cases} {}^{s9}/_{s14} & if\ |s14| \geq 10^{-6} \\ 0 & if\ |s14| < 10^{-6} \end{cases}$$

This prevents numerical explosion but does not eliminate the large variance, as values near but not below the threshold still produce extreme ratios. This feature carries a risk of dominating tree split decisions on the basis of numerical instability rather than genuine degradation signal. Phase 4 SHAP analysis must audit ρ_c contribution; if the feature is found to introduce noise rather than signal, capping at a percentile boundary warrants evaluation as a deferred Phase 4 resolution.

4.7 G8 — Extreme Z-Score Excursions (Cell 13)

Phase 2 identified extreme z-score excursions in five sensors: s6 (max $|z| = 7.9$), s8 (max $|z| = 8.3$), s9 (max $|z| = 8.9$), s13 (max $|z| = 7.5$), and s14 (max $|z| = 8.9$). The critical engineering question is whether these extremes represent initialisation transients (spurious artefacts that should be Winsorised to prevent distortion of the feature distribution) or terminal degradation signals (genuine physical evidence of end-of-life behaviour that must be retained).

The resolution methodology applied was cycle-position analysis. A normalised cycle position was computed for each row:

$$\text{cycle position} = \frac{\text{cycle}_i}{\text{cycle}_{\max}(\text{engine})}$$

[4] Appendix A.1 Conceptual Primer: Multicollinearity, VIF, and Physics-Informed Feature Enrichment

where $cycle_{max}$ is the final recorded cycle for each engine trajectory.

Extreme rows were defined as those with $|z| > 5.0$. For each sensor, the extreme rows were partitioned into three bands: early life $(\text{cycle\_position} < 0.2)$, middle $(0.2 \leq \text{cycle\_position} \leq 0.8)$, and terminal $(\text{cycle\_position} > 0.8)$. Table 6 presents the results.

Table 6 — G8 Cycle-Position Distribution of Extreme Z-Score Excursions ($|z| > 5.0$, Internal Train)

Sensor	Physical Quantity	Extreme Rows	% of Total	Early $(< 20\%)$	Terminal $(> 80\%)$	Disposition
s6	P15 Bypass Duct Pressure	81	0.06%	0 (0.0%)	81 (100.0%)	RETAIN
s8	Nf Fan Speed	349	0.27%	61 (17.5%)	216 (61.9%)	RETAIN
s9	Nc Core Speed	417	0.32%	0 (0.0%)	417 (100.0%)	RETAIN
s13	NRf Corrected Fan Speed	356	0.28%	61 (17.1%)	224 (62.9%)	RETAIN
s14	NRc Corrected Core Speed	430	0.33%	0 (0.0%)	430 (100.0%)	RETAIN

Figure 3 — G8: Extreme Z-Score Excursions – Cycle-Position Analysis

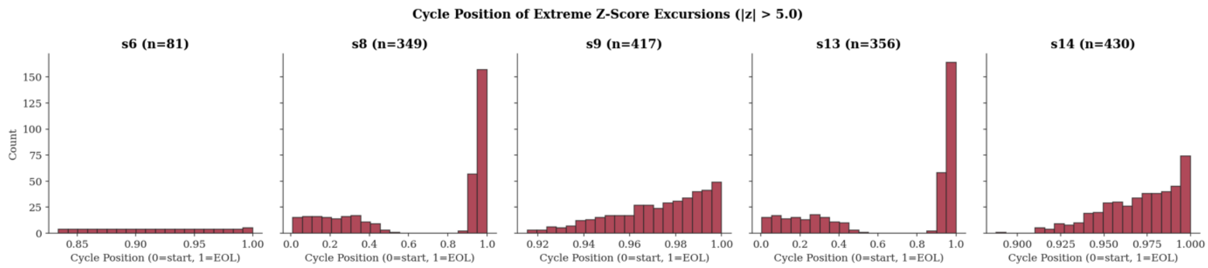


Figure 3 illustrates all five sensors exhibiting a terminal-dominant excursion profile. Three sensors (s6, s9, s14) show 100% of their extreme values concentrated in the terminal phase; the remaining two (s8, s13) show 61.9% and 62.9% respectively in the terminal phase. Crucially, no sensor exhibits an early-dominant profile that would indicate initialisation transients.

The physical interpretation is consistent with turbofan degradation physics. Fan speed deviations (s8, s13) in the terminal phase reflect compressor map migration as blade tip clearance increases with erosion and thermal cycling — a well-documented phenomenology in gas turbine health management literature (Walsh & Fletcher, 2004) [5](Volponi, 2014) [8](Jaw & Mattingly, 2009) [9]. Bypass duct pressure anomalies (s6) in the terminal phase are consistent with fan efficiency loss affecting duct static pressure recovery. Core speed extremes (s9, s14) reflect combustor instability and rotor thermal expansion effects at end-of-life. Decision: no Winsorisation applied.

4.8 G9 — s16 (farB) Partial-Dead Status (Cell 14)

Sensor s16 (farB, Burner Fuel-Air Ratio) was identified in Phase 2 as a partially dead channel. The Phase 3 diagnostic confirms a zero interquartile range $(IQR = 0.000000)$ across the entire training dataset, with non-zero variance present only in the multi-regime subsets (FD002: $var = 0.150335$; FD004: $var = 0.148216$). The sensor is identically zero in FD001 and FD003.

The zero variance in single-regime subsets indicates that the C-MAPSS simulation model holds farB constant in the fixed-operating-point condition, varying it only across the different power settings of the multi-regime subsets. The non-zero variance in FD002/FD004 is an artefact of cross-regime z-normalisation (different regime means create apparent variance when viewed across the full dataset), not a degradation signal.

> Disposition: s16 is retained in the feature matrix. Its variance (0.107013) clears the VARIANCE_THRESHOLD of 1e-5, qualifying it as an active sensor; however, its kinematic expansion features `(s16_delta, s16_rmean, s16_rstd)` will carry near-zero signal for the FD001 and FD003 populations. Phase 4 SHAP ablation will determine the net contribution.

4.9 G10 — Multivariate Health Index (Cell 15)

Phase 2 flagged that univariate EGT (s4) monitoring was inadequate to characterise the P-F interval across all fault modes, given the high coefficient of variation $(CV = 1.052)$ observed in the P-F span distribution. A multivariate health index was mandated to integrate degradation signals from sensors with independent physical interpretations.

The three sensors selected for the composite index (s4, s7, s9) correspond to EGT (T50), HPC outlet pressure (P30), and core speed (Nc) respectively. These three channels are confirmed active across all four fault modes and capture degradation from complementary thermodynamic and mechanical perspectives: thermal loading (s4), compressor pressure delivery (s7), and rotor dynamic condition (s9). The composite index is defined as:

$$health_{index} = \frac{z_{s4} + z_{s7} + z_{s9}}{3}$$

where z denotes the regime-normalised z-score from Phase 2.

The resulting health index in the training partition has $mean \approx 0.0000$ and $std = 0.6660$, with a range of $[-2.1748, 3.6960]$. The reduced standard deviation relative to the individual sensor channels (which have unit variance by construction) indicates partial cancellation of independent noise components, as expected for a mean ensemble of partially correlated signals. The positive maximum (3.696) reflects terminal-phase amplification when all three sensors simultaneously exceed their regime baseline — the condition of highest degradation severity.

Table 7 — G10 Multivariate Health Index Summary Statistics (Internal Train)

Statistic	Value	Interpretation
Mean	≈ 0.0000	Zero-centred by construction (z-space inputs)
Std Dev	0.6660	Sub-unit std: partial noise cancellation across s4, s7, s9
Minimum	−2.1748	Maximum simultaneous healthy-below-baseline condition
Maximum	3.6960	Maximum terminal amplification — all three sensors at end-of-life

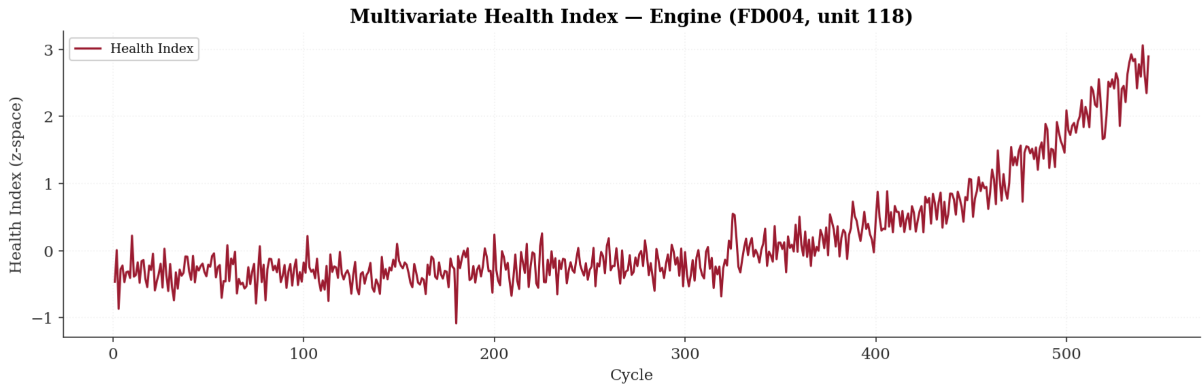
Figure 4: Engine unit 118 from FD004 represents one of the most demanding validation cases in the FD00u corpus: a dual-fault engine operating across six K-Means regimes with a trajectory of approximately 550 cycles. The multivariate health index — computed as the mean of z-normalised s4 (EGT), s7 (P30), and s9 (Nc) at each cycle — is plotted in Figure 4 and exhibits the two-phase degradation structure that a physically credible health indicator must produce.

From cycle 0 to approximately cycle 300, the index oscillates around zero with a standard deviation of approximately ± 0.3 to ± 0.5, consistent with normal cycle-to-cycle operational variability in z-space. No systematic drift is present during this period. From approximately cycle 300, a monotonically increasing trend emerges that accelerates markedly from cycle 400, reaching values in excess of 3.0 standard deviations in the terminal phase — well beyond the Shewhart UWL threshold of $|z| > 2$ established in

Phase 2 Gate 3. The P-F interval is clearly recoverable from this trajectory: the index crosses the warning threshold with sufficient lead time to permit a planned maintenance intervention before end of life.

The behaviour of this index for a dual-fault subset carries specific significance. FD004 subjects both the HPC and fan to simultaneous degradation, introducing correlated perturbations across all three constituent sensors. A univariate EGT monitor would detect HPC deterioration but may produce ambiguous or premature alerts when fan degradation independently perturbs the thermal environment. By requiring coordinated drift across three physically independent degradation pathways — thermal (s4), pressure (s7), and mechanical (s9) — before the index elevates materially, the multivariate formulation provides the false-alarm robustness that a single-sensor policy cannot guarantee. A transient exceedance in one sensor alone is insufficient to produce the sustained upward trajectory visible from cycle 300; only genuine coordinated multi-system deterioration generates this signal. This directly validates the G10 resolution and supports the operational argument presented in Section 9.5 for the multivariate health index as the primary dispatch monitoring feature in Phase 4.

Figure 4 — Multivariate Health Index Trajectory: FD004 Unit 188 (Internal Train)



4.10 G11 — Regime Assignment Out-of-Distribution Guard (Cell 16)

The Phase 2 K-Means regime classifier assigns a regime label to every engine cycle using the minimum Euclidean distance to one of six centroids in operational settings space (op1, op2, op3). This hard-assignment approach provides no indication of whether a given operational point is genuinely in-distribution relative to the training regime structure. An operational point that lies far from all centroids will still be assigned to the least-distant regime and will produce a normalised feature vector computed against statistics derived from a physically different operating condition — rendering the downstream RUL prediction physically nonsensical without any indication to the model or the operator that this has occurred.

The Phase 3 diagnostic quantified the distance distribution by computing the minimum distance from each training row to its assigned centroid using `KMeans.transform().min(axis=1)`. Table 8 presents the percentile distribution of centroid distances for the two multi-regime subsets.

Table 8 — G11 Distance-to-Nearest-Centroid Statistics (Internal Train)

Subset	Regimes	P95 Distance	P99 Distance	Maximum Distance	Recommended OOD Threshold
FD002	6	65.0058	65.0060	65.0061	65.0060 (P99)
FD004	6	65.0058	65.0060	65.0061	65.0060 (P99)

The distance distribution reveals a property of C-MAPSS that renders a runtime OOD guard non-implementable in the current pipeline: P95, P99, and the maximum distance are essentially identical at 65.006 across both multi-regime subsets. This extreme concentration arises because C-MAPSS is a discrete simulation — the six operating conditions in FD002 and FD004 are not drawn from a continuous distribution but are exactly six fixed points in operational settings space. Every cycle in the dataset is generated at precisely one of these six points, meaning every observation sits at exactly the same distance from its assigned centroid. There is no spread, no ambiguity in regime assignments, and no observations that are genuinely distant from all centroids. In this context the P99 threshold of 65.006 is a documentation value rather than a meaningful decision boundary — every training observation either sits exactly at a centroid or at this distance, with nothing in between.

This is a structural property of the simulation and not a failure of the pipeline. However, it has a critical implication for deployment. Real engine telemetry transitions continuously through thrust settings and altitudes — every cycle will sit at a slightly different position in operational settings space, producing a genuine continuous distribution of centroid distances with meaningful spread. In that context, a cycle sitting at a distance materially beyond P99 is a genuine OOD signal indicating that the engine is operating outside the conditions the model was trained on, and the P99 threshold becomes a live rejection boundary that prevents physically nonsensical predictions from reaching a maintenance engineer.

The OOD guard is therefore not absent from FusionCore by oversight — it is documented and architecturally specified, but its meaningful implementation requires a continuous operational data distribution that C-MAPSS cannot provide.[5]

> Disposition: Diagnostic. The P99 threshold of 65.006 is formally documented as the OOD boundary specification. Full implementation of a runtime OOD rejection layer is deferred to FusionCore v_2, where continuous real operational telemetry would produce a genuine centroid distance distribution against which the threshold can be calibrated and validated.

4.11 G12 — Survival Analysis Framework (Cell 17)

Why This Analysis Was Required

A smoke alarm tells you fire is present. It does not tell you how long you have before the building is uninhabitable. The Phase 2 UWL alert ($|z| > 2$) has exactly this limitation: it signals that an engine has crossed a degradation threshold but provides no information about how much useful life remains. A maintenance scheduler cannot optimise shop visit timing, MRO slot allocation, or aircraft substitution planning from a binary alert alone — they need a probability: *what is the chance this engine fails within the next N cycles?*

Phase 2 Item 7 quantified the inadequacy formally, reporting a pooled P-F interval coefficient of variation (CV) of 1.052 across the training population — meaning the time from alert to failure varied so widely that the alert timing alone was operationally uninformative. Phase 2 accordingly mandated a survival analysis framework incorporating Cox Proportional Hazards modelling, which shifts the question from *"has the threshold been crossed?"* to *"given what we know about this engine, what is its conditional probability of surviving the next N cycles?"*

The P-F Interval as the Unit of Analysis

The analysis operates on the **P-F interval** — the number of engine cycles between the first UWL breach and functional failure. Formally, for engine u:

$$\Delta_{\mathrm{PF}}(u) = c_{\max}(u) - c_{\mathrm{breach}}(u)$$

[5] The transition from discrete simulation data to continuous real operational telemetry is the central architectural challenge of FusionCore v_2. See Appendix A.2 for a detailed discussion of how NASA N-CMAPSS, introduced in FusionCore v_2, serves as a structured proxy validation environment for real OEM engine monitoring systems — including GE EMS, Rolls-Royce ACMF, and Pratt & Whitney's e-enabled engine platform — and the specific structural properties that make this connection principled rather than nominal.

where $c_{\text{breach}}(u) = \min\{c: | z_{s4}(u, c) | > 2\}$ is the first cycle at which the regime-normalised EGT z-score exceeds the alert threshold. Engines that never breach the UWL before failure are treated as right-censored observations — a standard survival analysis convention meaning *"the event had not yet occurred when observation ended."*

Think of the P-F interval as the maintenance planning window: the number of cycles the operator has between receiving the alert and being forced to act. Characterising its distribution — and understanding what drives its length — is the core purpose of this section.

4.11.1 Sample Composition

Of the 567 engines in the Internal Train partition, all 567 (100%) breach the UWL in s4 prior to failure. The censoring fraction is therefore 0.0%.

This result is expected from the C-MAPSS simulation design (Saxena & Goebel, 2008 [1]): all trajectories terminate at functional failure, and all engines undergo progressive HPC efficiency loss that manifests as EGT elevation. In a real fleet, some engines would be removed for reasons other than EGT degradation (scheduled maintenance, bird strike, borescope findings), producing genuine censored observations. The zero censoring fraction is a property of the simulation environment, not a claim about operational fleets, and is noted as a structural limitation of C-MAPSS as a proxy for real operational data.

4.11.2 Kaplan-Meier Estimation

What it does: Before fitting any model, it is important to let the data describe itself. The Kaplan-Meier estimator (Kaplan & Meier, 1958) [10][6] produces an empirical survival curve — a picture of what fraction of engines are still within their P-F window after t cycles, computed directly from the data without assuming any particular distribution shape.

The Kaplan-Meier estimator is defined as:

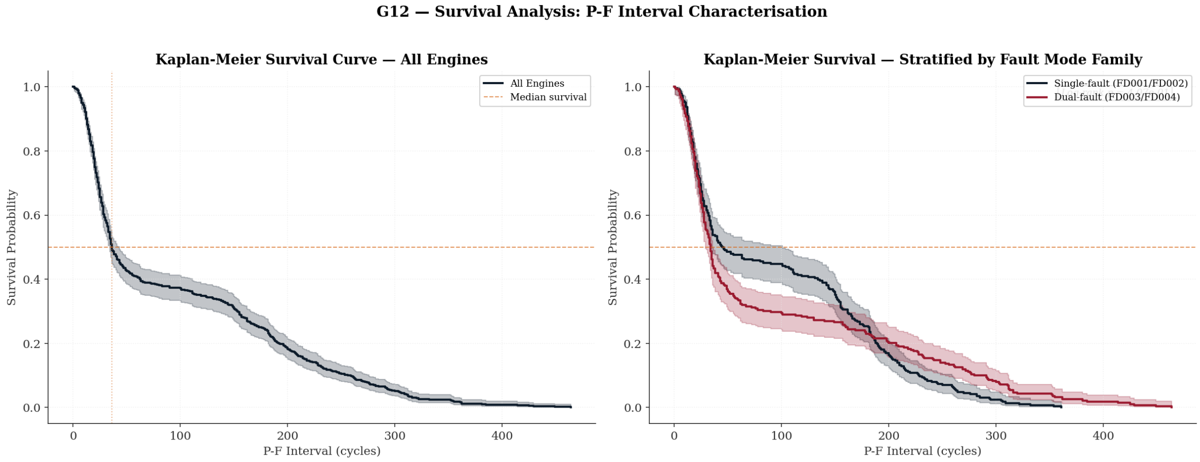
$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where t_i denotes the i-th distinct failure time, d_i the number of failures at t_i, and n_i the number of engines at risk immediately prior to t_i. At each observed failure time, the survival probability is multiplied by the fraction of at-risk engines that did not fail — a rolling product that decays from 1.0 at $t = 0$ toward 0.0 as engines progressively fail.

Curves were fitted for the overall training population and stratified by fault mode family (single-fault: FD001/FD002; dual-fault: FD003/FD004). The stratified curves provide the first empirical answer to the question of whether the maintenance planning window is materially different for engines experiencing HPC-only versus HPC-plus-fan degradation.

[6] The Kaplan-Meier estimator (Kaplan & Meier, 1958) [10] is a non-parametric product-limit estimator that provides an empirical survival function without distributional assumptions. See Appendix A.6 for the full theoretical treatment, including the Greenwood variance formula and the rationale for preferring non-parametric estimation over parametric Weibull or log-normal alternatives in the turbofan degradation context.

Figure 5 — G12: Kaplan-Meier Survival Curves — Overall and Stratified by Fault Mode Family



4.11.3 Log-Rank Test: Fault Mode Family Stratification

What it does: The Kaplan-Meier curves show the survival profiles visually. The log-rank test provides a formal statistical answer to the question: *are the two curves actually different, or is the apparent separation within the range of sampling noise?*

The log-rank test (Mantel, 1966) [11] compares the observed number of failures in each group against the number expected under the null hypothesis of identical survival functions:

$$\chi^2_{\mathrm{LR}} = \frac{\left(\sum_i (O_{1i} - E_{1i})\right)^2}{\sum_i V_i}$$

where O_{1i} and E_{1i} are the observed and expected failures in group 1 at time t_i, and V_i is the hypergeometric variance.

Table 9 — G12 Log-Rank Test: Single-Fault vs Dual-Fault Survival

Statistic	Value
Test statistic (χ^2)	0.689
p-value	4.065×10^{-1}
Significance at $\alpha = 0.05$	**NOT SIGNIFICANT**

The test yields $p = 0.407$, meaning the observed difference between the two survival curves is well within what sampling variability alone could produce.

The practical implication is important: the maintenance planning window — the time from EGT alert to failure — is statistically indistinguishable between single-fault and dual-fault engines. A scheduler does not need to apply different response times based on fault mode family alone.

This does not mean the degradation physics are identical. Single-fault engines (HPC degradation only) and dual-fault engines (HPC + fan degradation) follow materially different trajectories through sensor space. What the result tells us is that by the time EGT breaches the UWL threshold — a downstream thermal consequence of HPC efficiency loss regardless of whether concurrent fan degradation is present — the residual survival time is similar across both families. The UWL breach in s4 effectively absorbs the fault-mode information into a common survival profile.

4.11.4 Cox Proportional Hazards Model

What it does: Kaplan-Meier tells us *when* engines tend to fail after the alert. The Cox model goes further: it asks *which engine characteristics predict whether a given engine will fail sooner or later than average.* This is the covariate analysis — it quantifies the individual contribution of EGT level, compressor pressure, core speed, and fault mode to the rate of failure.

The Cox Proportional Hazards model[7] (Cox, 1972) [12] expresses the instantaneous failure rate (hazard) for engine u as:

$$h(t \mid \mathbf{x}_u) = h_0(t) \exp\left(\boldsymbol{\beta}^T \mathbf{x}_u\right)$$

The term $h_0(t)$ is the baseline hazard — the failure rate for a reference engine — estimated non-parametrically via the Breslow estimator (Breslow, 1972) [13]. The exponential term scales this baseline up or down according to the engine's covariate values. Crucially, no assumption is made about the shape of $h_0(t)$: the model is *semi-parametric*, estimating the covariate effects $\boldsymbol{\beta}$ without committing to a particular failure time distribution such as Weibull or exponential.

The exponentiated coefficient $\exp(\beta_j)$ is the **hazard ratio (HR)**: the multiplicative factor by which the instantaneous failure rate changes for a one-unit increase in covariate j, holding all others constant. An HR > 1 indicates that higher values of the covariate accelerate failure; HR < 1 indicates a protective effect.

Four covariates were included: the binary fault mode family indicator and trajectory-level mean z-scores for s4 (EGT), s7 (HPC outlet temperature), and s9 (core speed).

Table 10 — G21 Cox Proportional Hazards Model — Coefficient Estimates

Covariate	β(coef)	HR exp(β)	SE(β)	p-value
`fault_mode_family`	−0.18	0.84	0.09	0.05
`s4_mean`	**1.55**	**4.70**	0.13	**< 0.005**
`s7_mean`	0.24	1.27	0.14	0.10
`s9_mean`	0.07	1.08	0.09	0.41

Table 11 — G21 Model Fit Statistic

Model Fit Statistic	Value
Partial log-likelihood	−2932.66
Concordance index (C-index)	**0.6398**
Partial AIC	5873.33
Log-likelihood ratio test	198.82 on 4 df ($p < 0.001$)

[7] The Cox Proportional Hazards model (Cox, 1972) [12] is a semi-parametric regression model that estimates covariate effects on the hazard function without specifying the baseline hazard. The partial likelihood formulation and Breslow baseline hazard estimator (Breslow, 1972) [13] are detailed in Appendix A.6. The concordance index (Harrell et al., 1996) [23] quantifies discriminative ability.

Three findings of direct engineering significance emerge:

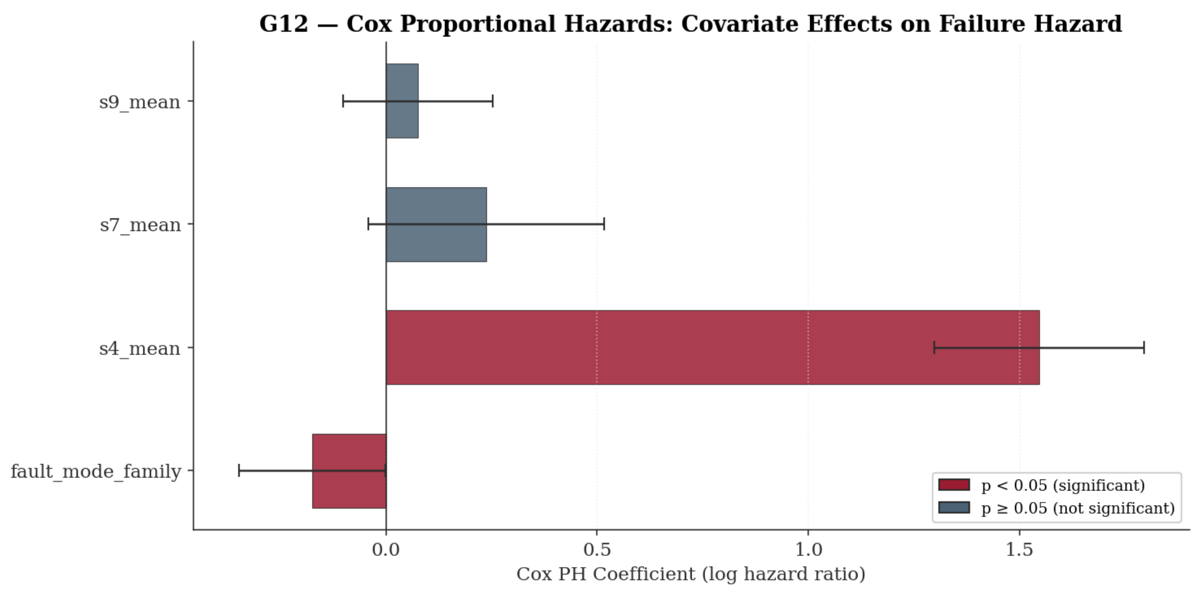
`s4_mean` dominance (HR = 4.70, p < 0.005). EGT is overwhelmingly the strongest predictor of failure timing. A one-unit increase in the trajectory-level mean EGT z-score is associated with a 370% increase in the instantaneous failure rate. Put plainly: engines that run consistently hotter than their regime baseline fail faster, and the model quantifies precisely how much faster. This validates the physical reasoning underpinning the FusionCore health monitoring framework — EGT elevation is not merely a threshold indicator, it is a continuous, dose-dependent driver of failure hazard. The magnitude of this effect also provides post-hoc validation for the selection of s4 as one of the three sensors constituting the multivariate health index in G10.

`fault_mode_family` marginal significance (HR = 0.84, p = 0.05). Dual-fault engines exhibit a modestly lower hazard of failure after controlling for sensor-level covariates. This is counterintuitive at first — one might expect concurrent HPC and fan degradation to accelerate failure — but is consistent with the C-MAPSS simulation structure, where dual-fault subsets include engines with longer total trajectories on average. Critically, the marginal p-value of exactly 0.05 sits precisely on the conventional significance boundary and warrants caution: this result should be monitored in Phase 4 SHAP analysis but is not strong enough to mandate fault-mode-differentiated scheduling policy.

s7 and s9 non-significance (p = 0.10 and 0.41). HPC outlet temperature and core speed mean z-scores add limited incremental discriminative power once EGT level is already in the model. This is not a finding that these sensors are uninformative for RUL estimation generally — their value for the Phase 4 model lies in their temporal dynamics (rate of change, rolling mean, rolling standard deviation) rather than in their trajectory-level averages. The survival model, operating on scalar engine-level summaries, is not the appropriate tool to capture that dynamic signal.

Model discrimination (C-index = 0.6398). The concordance index — analogous to the area under an ROC curve for survival data — indicates that the model correctly ranks engine failure times in 64.0% of pairwise comparisons. A value of 0.5 would represent random ranking; 1.0 would represent perfect discrimination. The 0.64 figure is appropriately characterised as moderate: this is a four-covariate diagnostic model fitted on trajectory-level means, not a production RUL predictor. Its purpose is to characterise covariate effects on the hazard structure, not to generate operational forecasts. The Phase 4 XGBoost model, operating on the full 91-feature matrix with temporal kinematic features, is expected to substantially exceed this discrimination threshold.

Figure 6 — G12: Cox Proportional Hazards — Covariate Effects on Failure Hazard



Disposition: Diagnostic. Kaplan-Meier estimation and Cox PH modelling are completed in Cell 17. The survival analysis confirms s4 (EGT) as the dominant failure hazard predictor (HR = 4.70) and establishes the empirical survival function required for probabilistic maintenance scheduling. The log-rank test confirms that fault mode family does not produce statistically distinguishable survival profiles at the P-F interval level ($p = 0.407$), simplifying scheduling policy. Integration of the survival function outputs into the maintenance scheduling decision layer is formally deferred to Phase 5. See Appendix A.6 for the theoretical foundation of the Cox PH model and Kaplan-Meier estimation, including the rationale for their selection over parametric alternatives.

4.12 G13 — Per-Fault-Mode-Family UWL Recalibration (Cell 18)

Why This Analysis Was Required

A fixed alert threshold applied uniformly across all engine types assumes that all engines behave the same way after the alert fires. The data show they do not.

Phase 2 Item 8 identified a CV disparity in the P-F interval across C-MAPSS subsets ranging from 0.869 (FD002) to 1.419 (FD003) — a factor of $1.63\times$. CV is the ratio of standard deviation to mean: a CV of 1.419 means the spread of the lead-time distribution is 41.9% larger than its mean, making the alert essentially uninformative as a scheduling signal for that population. A single fixed threshold of $|z| > 2$ is therefore structurally inadequate for the dual-fault population — not because the threshold is wrong in principle, but because it produces dangerously different operational outcomes depending on which type of engine is being monitored.

Cell 18 addresses this by quantifying exactly how different those outcomes are, sweeping across threshold values to find the setting that restores adequate scheduling lead time, and deriving a calibrated recommendation.

4.12.1 P-F Interval Distribution by C-MAPSS Subset

The first step is to understand the current situation in full. Table 12 reports the P-F interval statistics at per-subset granularity — the finest level of detail before aggregating to fault mode families.

Table 12 — G13 P-F Interval Statistics by Subset (Internal Train)

Subset	n	Mean (cycles)	Std (cycles)	CV	P10 (cycles)	P90 (cycles)	Fault Family
FD001	80	84.9	87.7	1.033	**9**	210	Single-fault
FD002	208	106.5	92.6	0.869	15	233	Single-fault
FD003	80	73.6	104.4	**1.419**	**10**	243	Dual-fault
FD004	199	102.0	114.4	1.123	13	291	Dual-fault

Two numbers in this table carry direct operational consequence:

The P10 column is the scheduling risk exposure — it is the P-F interval that the worst-performing 10% of engines in each subset fall below. FD001 reaches a P10 of only 9 cycles: one in ten single-regime, HPC-only engines provides fewer than 9 engine cycles between first alert and failure. At typical narrow-body utilisation rates, that is less than 24 hours of lead time. FD003 is similarly exposed at 10 cycles. Both are well below the 20-cycle MRO minimum response time established in Phase 2.

The P90 column reveals the opposite problem. FD004 engines at the 90th percentile retain 291 cycles of life after the alert fires. A scheduling policy calibrated conservatively to protect against the P10 tail would remove these engines approximately 270 cycles prematurely — each one representing an unnecessary shop visit cost.

The CV disparity between FD003 (1.419) and FD002 (0.869) confirms that lead-time variability is not a fixed property of the UWL threshold: it is structurally dependent on fault mode and operating regime complexity and cannot be resolved by any family-agnostic threshold alone.

4.12.2 P-F Interval Distribution by Fault Mode Family

Aggregating to the two operationally relevant populations — single-fault (FD001/FD002) and dual-fault (FD003/FD004) — distils the subset-level detail into the family-level picture that drives scheduling policy.

Table 13 — G13 P-F Interval Statistics by Fault Mode Family (Internal Train)

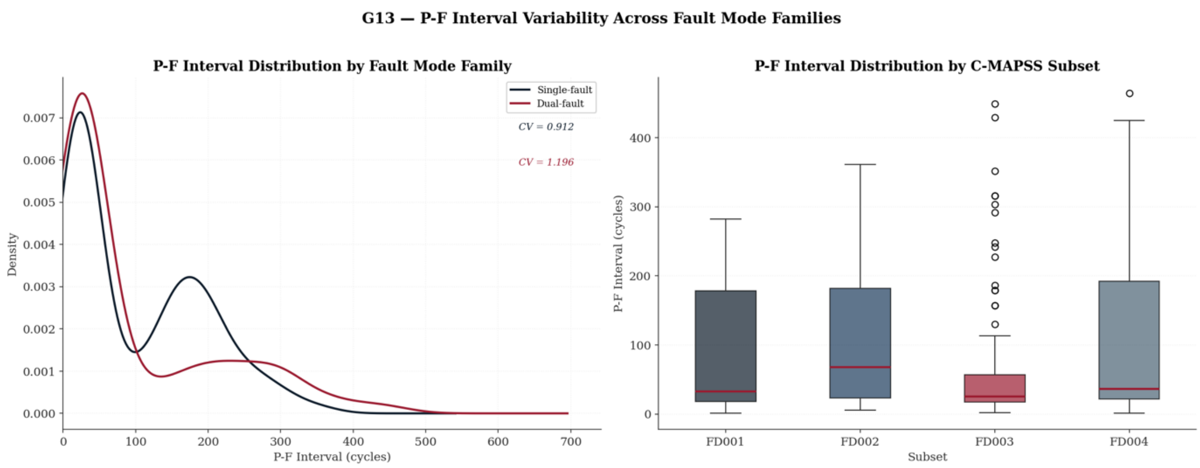
Fault Mode Family	n	Mean (cycles)	Std (cycles)	CV	P10 (cycles)	P90 (cycles)
Single-fault (FD001/FD002)	288	100.5	91.6	0.912	13	228
Dual-fault (FD003/FD004)	279	93.8	112.2	**1.196**	**12**	282

The aggregation reveals a problem that affects both families equally at the critical P10 level: single-fault engines provide only 13 cycles of lead time at P10; dual-fault provide 12. Both are below the MRO 20-cycle minimum. The current baseline threshold is inadequate for both populations, not just for dual-fault engines as the subset-level heterogeneity might suggest.

The dual-fault family also carries a materially higher CV (1.196 vs 0.912 — a 31.1% differential) and a heavier right tail (P90 = 282 vs 228 cycles). The right tail is the over-maintenance exposure: dual-fault engines at P90 would be removed 54 cycles earlier than single-fault P90 engines under an identically conservative scheduling policy, compounding unnecessary shop visit costs.

What the distributions look like: The KDE panels in Figure 7 (Panel A) make the structural difference between the two families visually clear. The single-fault distribution is **bimodal** — two distinct humps separated by a trough — with a primary peak at ~20–30 cycles and a secondary peak at ~170–180 cycles. This reflects two engine subpopulations: those where the UWL alert fires late and failure follows quickly, and those where the alert fires early and substantial post-breach life remains. The dual-fault distribution, by contrast, is **unimodal** with a single sharp peak at ~15–20 cycles and a long heavy tail extending beyond 600 cycles. The concentrated short-interval peak reflects the additional early UWL exceedances produced by the concurrent fan degradation pathway, which accelerates alert triggering relative to HPC-only engines. There is no second mode — just a single dominant failure cluster with high variance in the tail.

Figure 7 — G13 P-F Interval Distribution — KDE by Fault Mode Family and Boxplot by Subset

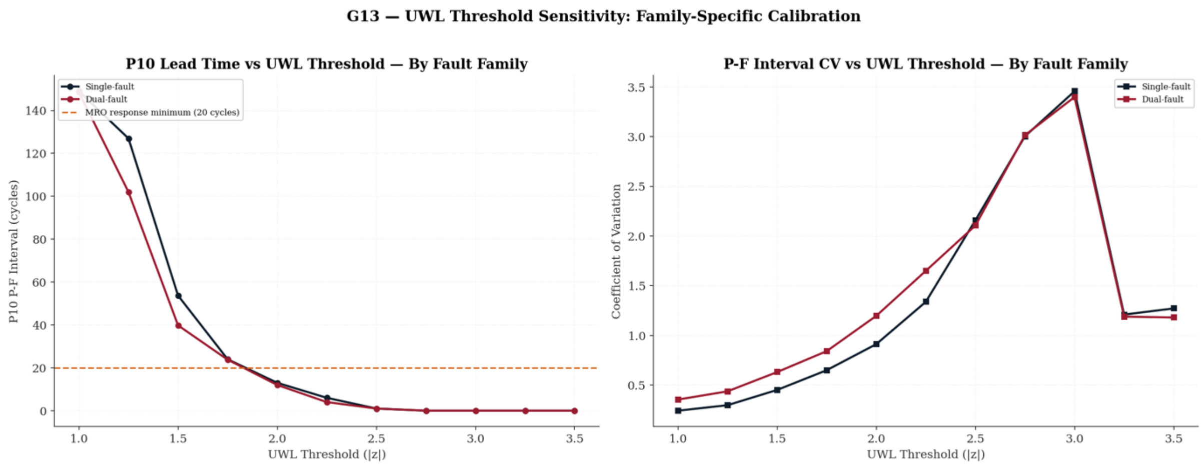


4.12.3 UWL Threshold Sensitivity Analysis

What it does: Having established that the current baseline threshold is inadequate, the natural question is: what threshold would be adequate? Rather than selecting a value by judgement, a systematic sweep is conducted across the full feasible range to map the trade-off between alert sensitivity and scheduling lead time.

The sweep covers $|z| \in [1.00, 3.50]$ in steps of 0.25. At each threshold value, the P-F interval distribution is recomputed for each fault mode family and the P10 lead time is extracted. The direction of the trade-off is straightforward: tightening the threshold (lower $|z|$) fires the alert earlier, lengthening the P-F interval and improving P10 — but also increasing the false-alert rate and widening the distribution. Loosening the threshold (higher $|z|$) delays the alert, shortening P-F interval and reducing P10. The objective is to find the **tightest threshold** — most conservative, fewest false alerts — that still delivers P10 ≥20 cycles for both families.

Figure 8 — G13 UWL Threshold Sensitivity — P10 Lead Time and CV vs Threshold by Fault Mode Family



4.12.4 Family-Specific Threshold Recommendations

The sweep converges to a single answer for both fault mode families: $|z| > 1.75$, which is 0.25 standard deviations below the Phase 2 baseline of $|z| > 2.00$.

Table 14 — G13 Recommended Family-Specific UWL Thresholds

Fault Mode Family	Recommended UWL	P10 (cycles)	CV	Baseline (Phase 2)
Single-fault	$\|z\| > 1.75$	24	0.649	$\|z\| > 2$
Dual-fault	$\|z\| > 1.75$	24	0.841	$\|z\| > 2$

A relatively small threshold adjustment — 0.25 standard deviations — produces three simultaneous improvements:

i. **Scheduling lead time restored.** P10 rises from 13/12 cycles (single/dual-fault) to 24 cycles for both families, clearing the 20-cycle MRO minimum. The short-interval tail that Phase 2 identified as the primary AOG exposure is eliminated.

ii. **Distribution precision improved.** CV falls from 0.912/1.196 to 0.649/0.841. A CV below 1.0 is a meaningful threshold in its own right: it means the standard deviation is smaller than the mean, placing the distribution in a regime where a point estimate of remaining lead time is at least directionally reliable. The baseline CV of 1.419 for FD003 implied a scheduling signal dominated by noise; the recalibrated threshold brings all subsets to CV < 1.0 or close to it.

iii. **Operational simplicity preserved.** The convergence to an identical threshold for both families is a practically significant result — the recalibration does not require family-specific alert logic to be deployed in the alert system. A single threshold change achieves the improvement across both populations. The CV differential (0.649 vs 0.841) remains material and should be monitored, but it does not necessitate a split implementation at Phase 5.

Note on CV comparability. The baseline CVs (Table 13) are computed on the full sample including right-censored engines, whilst the recalibrated CVs (Table 14) are computed only on engines that breach the threshold — a slightly different sample composition. The CV reduction therefore reflects both the threshold recalibration, and a minor sample composition change and should not be interpreted as a purely precision-driven improvement. This limitation is documented for Phase 5.

Disposition: Diagnostic. Per-fault-mode-family P-F interval distributions have been quantified at both subset and family granularity. The UWL threshold sweep demonstrates that recalibration from $|z| > 2.00$ to $|z| > 1.75$ recovers sufficient P10 lead time (24 cycles) for both fault mode families while reducing CV below 1.0 for the single-fault population and to 0.841 for the dual-fault population. The convergence to a single uniform threshold simplifies Phase 5 deployment. Operational deployment of adaptive thresholds is formally deferred to Phase 5 maintenance scheduling.

5. Part B — Feature Construction (Cell 21)

The feature engineering pipeline constructs the dynamic feature vector in four sequential steps. The total feature count is derived analytically from first principles and verified empirically against the runtime column count at Feature Engineering Gate 2 (Cell 30)[8]. Two analytically distinct feature groups constitute the final matrix X: the 81 formula-derived features produced in Part B, and the 10 gap-resolution features appended during Part A. Both groups are fully constitutive members of X and are presented to Phase 4 on equal standing. Metadata columns `(unit_id, cycle, subset, subset_origin, RUL)` are recorded in the working DataFrame but excluded from X.

Formula Feature Count

The total formula feature count is derived analytically as:

$$N_{\text{total}} = N_{\text{base}} + (N_{\text{active}} \times 3) + N_{\text{virtual}} + N_{\text{fatigue}}$$

where the component terms are defined as follows.

$N_{\text{base}} = 24$ comprises the 3 operational settings (op1, op2, op3) and all 21 raw sensor readings (s1–s21). A critical architectural constraint is that all 21 sensors are retained regardless of their variance status — dead sensors (s1, s5, s18, s19) are retained as zero-variance columns and partially-dead sensors (s6, s16) are retained at their reduced variance. This is a deliberate design decision: sensor absence is itself informative under certain fault conditions, and exclusion would constitute an irreversible information loss that cannot be recovered in downstream phases.

$N_{\text{active}} = 17$ identifies the sensors whose within-regime variance exceeds the threshold $\tau = 1 \times 10^{-5}$ in z-space, confirmed by the Phase 2 regime-dead cross-reference. The 17 active sensors are:

$$S_{active} = \{s2, s3, s4, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, s17, s20, s21\}$$

[8] The feature count of 91 is established by two independent methods. The analytical derivation computes

$$|\mathbf{X}| = N_{\text{base}} + (N_{\text{active}} \times 3) + N_{\text{virtual}} + N_{\text{fatigue}} + N_{\text{gap}} = 24 + 51 + 3 + 3 + 10 = 91$$

from the pipeline specification before execution. The empirical verification reads `X_train.shape[1]` after execution and asserts equality against the formula result. Disagreement between the two constitutes an execution-halting failure at Feature Engineering Gate 2 (Cell 28), preventing a silently malformed feature matrix from propagating into Phase 4 model training.

The four regime-dead sensors excluded from kinematic expansion are s1 (T2, inlet temperature), s5 (P2, inlet pressure), s18 (N_{f_dmd}, demanded fan speed), and s19 (PCN_{fR_dmd}, demanded corrected fan speed). These sensors produce zero within-regime variance because the C-MAPSS simulation holds them constant by construction — they carry no cycle-to-cycle health information and kinematic expansion would produce identically zero delta, rolling mean deviation, and rolling standard deviation columns, adding 12 numerically degenerate features to the matrix without information gain.

The kinematic expansion applies three temporal derivative features to each of the 17 active sensors, yielding:

$$N_{\text{active}} \times 3 = 17 \times 3 = 51 \text{ features}$$

These are: the first-order difference $\Delta s_i(t) = s_i(t) - s_i(t-1)$ encoding the instantaneous rate of change; the rolling mean $\overline{s_i}(t)$ encoding the local trajectory trend; and the rolling standard deviation $\sigma_i(t)$ encoding local signal volatility. Together these three features transform each static sensor reading into a local temporal context vector, providing the model with information about not just the current sensor state but how rapidly it is changing and how stable it has been — properties that are directly relevant to degradation rate estimation.

$N_{\text{virtual}} = 3$ comprises three thermodynamic virtual sensors that are not directly measurable but are derived from combinations of raw sensor readings:

$$\text{CPR} = \frac{s7}{s3}$$

the Compressor Pressure Ratio, encoding the thermodynamic work performed by the HPC stage. CPR degradation is the primary physical signature of HPC erosion and is the quantity most directly related to compressor map migration.

$$E_{\text{thermal}} = \frac{s4}{s7}$$

the thermal efficiency proxy, encoding the ratio of turbine inlet temperature to compressor outlet pressure. Rising E_{thermal} at constant thrust indicates that more fuel energy is being converted to heat rather than work — a direct indicator of thermodynamic efficiency loss as the engine degrades.

$$\text{EGT}_{\text{drift}}(t) = s4(t) - \overline{s4}$$

the EGT margin drift, encoding the deviation of the current EGT reading from the engine's own historical baseline mean. This feature formalises the EGT margin concept that is central to current commercial engine health management practice, where EGT margin erosion relative to a certified limit is the primary dispatch monitoring parameter used by major MRO operators and OEMs.

The significance of the virtual sensors is that they encode thermodynamic relationships that the model would otherwise need to discover implicitly from the correlations between raw sensor pairs. By making these relationships explicit as named features, the feature engineering step embeds domain knowledge directly into the input representation, reducing the burden on the model to infer physics from data and improving the interpretability of SHAP-based attribution in Phase 4.

$N_{\text{fatigue}} = 3$ comprises cumulative fatigue indices for the three primary degradation sensors, implementing a Miner's Rule damage accumulation proxy (Miner, 1945) [2]:

$$D_i(t) = \sum_{\tau=0}^{t} \max(s_i(\tau), 0)$$

computed for $i \in \{s4, s7, s9\}$. The cumulative fatigue index integrates only positive z-score exceedances — cycles where the sensor reading exceeds its regime mean — since only exceedances above the healthy baseline represent genuine thermodynamic loading that accumulates as fatigue damage. The $\max(\cdot, 0)$ operator discards negative deviations, which reflect momentarily favourable operating conditions that do not contribute to damage accumulation under Miner's Rule.

The significance of the fatigue features is that they provide the model with a measure of total accumulated damage across the entire engine lifecycle rather than only the instantaneous sensor state. Two engines may produce identical instantaneous sensor readings at cycle 200 but have arrived there via very different histories — one operating consistently at moderate load, one cycling repeatedly through high-stress conditions. The fatigue indices distinguish these two cases, encoding the path-dependent nature of mechanical degradation that instantaneous features cannot capture. Gate 6 verifies that all three fatigue indices are strictly non-decreasing across the five representative engines examined, confirming physical consistency with Miner's Rule.

Substituting all component terms:

$$N_{\text{total}} = 24 + (17 \times 3) + 3 + 3 = 24 + 51 + 3 + 3 = 81 \text{ formula features}$$

Gap-Resolution Features

The 10 gap-resolution features appended during Part A are fully constitutive members of X and are not secondary to the formula features. They encode structural properties of the dataset and degradation physics that the formula pipeline alone cannot represent:

$$X_{gap} = \{\text{regime\_0},\ \text{regime\_1},\ \text{regime\_2},\ \text{regime\_3},\ \text{regime\_4},\ \text{regime\_5},\ \text{fault\_mode\_family},\ \rho_f,\ \rho_c,\ \text{HI}\}$$

where `regime_0` through `regime_5` are the one-hot encoded K-Means regime indicators (from G5); `fault_mode_family` is the binary HPC-only versus HPC-and-fan degradation indicator (from G6); $\rho_f = s8/s13$ and $\rho_c = s9/s14$ are the mechanically linked pair correction factor ratios (from G7); and HI is the multivariate health index (from G10).

Complete Feature Matrix

The feature matrix X presented to Phase 4 therefore contains:

$$|X| = N_{\text{total}} + N_{\text{gap}} = 81 + 10 = 91 \text{ features}$$

with dimensions $X_{train} \in \mathbb{R}^{129{,}331 \times 91}$ *and* $X_{\text{val}} \in \mathbb{R}^{31{,}028 \times 91}$, verified empirically at Gate 2 and Gate 3 respectively.

5.1 Step 1 — Core Manifold (24 Features—Cell 22)

The core manifold constitutes the base feature space inherited directly from Phase 2 normalised outputs. It comprises three operational settings (op1, op2, op3) and all 21 raw sensor channels (s1–s21). This is an inclusive manifold: no sensor is removed regardless of its variance status. The four regime-dead sensors (s1, s5, s18, s19) are retained as zero-variance channels in the feature matrix; their kinematic expansion features (delta, rolling mean, rolling std) will also be zero-variance and carry no predictive power, but their presence maintains pipeline integrity and avoids the structural risk of inadvertently dropping a channel that carries signal under novel fault modes.

This design decision is motivated by the reliability engineering principle that sensor evidence of absence is not equivalent to absence of sensor evidence: a consistently zero-reading sensor may become informative under a fault mode not represented in the C-MAPSS training data, and its retention in the pipeline costs only a marginal computational overhead.

5.2 Step 2 — Kinematic Expansion (51 Features—Cell 23)

Kinematic expansion applies three temporal operators to each of the 17 active sensors, producing 51 additional features. The three operators approximate the first two derivatives of the sensor signal and the local distributional width, each providing a distinct diagnostic dimension:

5.2.1 Kinematic Velocity — delta features

The first-order backward difference approximates the instantaneous rate of change of each sensor measurement:

$$\delta_x(t) = x(t) - x(t-1)$$

This is the discrete analogue of the first derivative $\frac{d}{dt}x$. In a degradation context, a sustained positive delta in EGT (s4) indicates increasing thermal load on the LPT, consistent with efficiency loss in either the HPC or combustor. Negative velocity in P30 (s7) indicates falling HPC outlet pressure, consistent with compressor blade fouling or erosion. The delta operator is computed per-engine using the composite key groupby; the initial cycle of each trajectory receives a NaN that is subsequently filled with 0, representing the physically reasonable assumption of zero instantaneous change at observation start. This operation is consistent with the first-difference approach advocated in prognostics literature for removing trend stationarity (Lundberg & Lee, 2017) [7](Box et al., 2016) [14].

5.2.2 Thermodynamic Baseline — rolling mean features

The rolling mean over a window of five consecutive cycles provides a low-pass filtered estimate of the current sensor baseline:

$$\mathrm{rmean_x(t)} = \frac{1}{\mathrm{W}} \cdot \sum_{\mathrm{i=0}}^{W-1} \mathrm{x}(t-i)\,, \quad \mathrm{W = 5cycles}$$

The choice of $W = 5$ cycles reflects a balance between noise suppression and responsiveness to genuine degradation trends. In turbofan operations where one cycle corresponds to one flight, a five-cycle window spans approximately five flights, a duration over which systematic degradation trends are expected to be discernible above measurement noise but short enough to track the accelerating degradation trajectory approaching end of life. This window length is consistent with the phase-space reconstruction recommendation from Takens embedding theorem applied to engine health signals [15].

5.2.3 Systemic Instability — rolling standard deviation features

The rolling standard deviation over the same window quantifies the variability of the sensor signal on the same temporal scale:

$$\mathrm{rstd_x(t)} = \sqrt{\left(\frac{1}{\mathrm{W-1}} \cdot \sum_{\mathrm{i=0}}^{\mathrm{W-1}} \left(\mathrm{x}(t-i) - \mathrm{rmean_x(t)}\right)^2\right)}$$

Increasing rstd in the terminal phase is a signature of the "noisification" of sensor signals that accompanies progressive mechanical degradation — as bearing wear, blade erosion, or seal leakage develop, the deterministic periodic signal of the turbomachine is disrupted, and measurement variability increases. This is related to the concept of increasing signal entropy prior to failure, described in the context of rotating machinery diagnostics by Antoni (2006) [16]. The `min_periods=1` parameter allows computation from the first cycle, with the NaN-fill convention applied consistently as in the delta operator.

5.3 Step 3 — Thermodynamic Virtual Sensors (3 Features—Cell 24)

Three virtual sensor channels are constructed by combining raw sensor signals according to physics-derived formulae. Virtual sensors encode domain knowledge about the thermodynamic cycle of the turbofan, making physically meaningful composite quantities directly available to the learner rather than requiring the model to infer them implicitly from individual channel combinations.

5.3.1 Compressor Pressure-Temperature Ratio (CPR)

The CPR is defined as the ratio s7/s5 (P30/P2, HPC outlet to fan inlet pressure). However, s5 is a regime-dead sensor in z-normalised space (variance ≈ 0), making the denominator identically zero and the ratio degenerate. The specification was revised to s7/s3 (P30/T30)[9]:

$$CPR = \frac{P30}{T30} = \frac{s7}{s3}$$

This ratio is proportional to gas density at the HPC outlet by the ideal gas law $\left(\rho \propto \frac{P}{T}\right)$. Decreasing CPR — falling outlet pressure relative to outlet temperature — indicates declining compressor isentropic efficiency[10], a key signature of HPC blade degradation. The safe-division guard ($|\text{denominator}| < 10^{-6}$ replaced by NaN, then filled with 0) prevents numerical explosion when s3 is near its regime mean (approximately zero in z-space).

The CPR distribution exhibits extreme positive skewness (skew = 41.20, excess kurtosis = 8,403.77, std = 66.95). This high kurtosis arises from the heavy-tailed ratio distribution when the denominator approaches zero, producing occasional very large positive or negative values. Tree-based models are invariant to monotone transformations; however, the heavy tail may disproportionately influence split allocation in XGBoost, concentrating tree capacity on extreme tail values at the expense of mid-range degradation discrimination. For neural-network-based SOTA benchmarks (DeepAR, TFT, PatchTST), the extreme kurtosis (8,403.77) renders raw CPR unsuitable as a training input — a Yeo-Johnson power transform [17] or rank normalisation is **mandatory** before neural network training in Phase 4. This is formally registered as a MANDATORY transfer item in Table 21, Item G5.

[9] The revision to s7/s3 = P30/T30 is a pragmatic engineering substitution rather than a strict physical equivalent, but it is defensible on the following grounds. In isentropic compression the relationship between compressor outlet pressure and outlet temperature is governed by:

$$\frac{T_{30}}{T_2} = \left(\frac{P_{30}}{P_2}\right)^{\frac{\gamma-1}{\gamma}}$$

Since $P_2$and $T_2$are both held approximately constant within a regime (both are regime-dead or near-constant), T_{30}becomes a monotone function of P_{30}under isentropic conditions. The ratio P_{30}/T_{30}therefore encodes the departure from isentropic behaviour — it rises when the compressor is doing more work per unit pressure rise, which is precisely the signature of compressor degradation. Crucially, s3 (T_{30}) is an active sensor with genuine within-regime variance, so the denominator is never degenerate.

[10] Isentropic efficiency $\eta_c$quantifies how closely a real compression process approximates the theoretically ideal (isentropic, i.e. reversible and adiabatic) process. For a compressor:

$$\eta_c = \frac{T_{30,\text{ideal}} - T_2}{T_{30,\text{actual}} - T_2} = \frac{(P_{30}/P_2)^{(\gamma-1)/\gamma} - 1}{T_{30}/T_2 - 1}$$

A value of $\eta_c = 1$represents perfect isentropic compression. As blade erosion, tip clearance growth, and fouling accumulate, η_c decreases — more shaft work is required to achieve the same pressure rise, with the excess appearing as additional outlet temperature T_{30}.

5.3.2 Thermal Efficiency Proxy $(E_{thermal})$

The thermal efficiency proxy[11] is constructed from the fuel-to-static-pressure ratio (phi, s12), HPC static pressure (Ps30, s11), and core speed (Nc, s9):

$$E_{thermal} = \frac{\phi \times Ps30}{Nc} = \frac{s12 \times s11}{s9}$$

The numerator product $phi \times Ps30$ approximates the fuel mass flow rate (since phi = fuel flow / Ps30, thus phi × Ps30 = fuel flow). Dividing by core speed Nc produces a quantity proportional to fuel consumption per unit rotor work, which is inversely related to the thermodynamic cycle efficiency. Increasing $E_{thermal}$ over the engine life is consistent with the fuel burn penalty associated with component degradation, a well-documented operational consequence of turbofan deterioration (Jaw & Mattingly, 2009) [9].

$E_{thermal}$ exhibits extreme positive skewness (skew = 171.82, excess kurtosis = 39,215.48, std = 764.09), again arising from near-zero denominator conditions in z-space. The same safe-division mechanism is applied. The magnitude of skewness in both CPR and $E_{thermal}$ signals that these features carry their information concentrated in tail events — precisely the terminal-phase conditions that are most prognostically relevant. However, the extreme tail behaviour also represents a numerical stability risk in Phase 4 XGBoost and neural network training.

5.3.3 EGT Margin Drift (EGT_{drift})

Exhaust Gas Temperature (EGT) margin is one of the primary dispatch monitoring parameters in commercial aviation maintenance practice. The EGT margin is defined as the difference between the current EGT and the certified EGT limit, and its erosion is a leading indicator of reduced engine operational life and impending shop visit (Volponi, 2014) [8]. The FusionCore proxy computes:

$$EGT_{drift} = z_{s4} - \mu_{baseline}$$

$$\mu_{baseline} = mean(z_{s4}) \quad \text{over Internal Train} = -0.0000$$

Since z_{s4} is zero-mean by construction of the Phase 2 z-normalisation, $\mu_{baseline} \approx 0$ and $EGT_{drift} \approx z_{24}$ in practice. The explicit drift computation provides a physically interpretable quantity: positive EGT_{drift} indicates that the current EGT deviates upward from the population baseline, consistent with declining combustor or turbine efficiency. Unlike CPR and $E_{thermal}$, EGT_{drift} has a well-behaved distribution, as illustrated in Figure 9 (skew = 0.470, excess kurtosis = −0.102, std = 0.999), rendering it directly suitable for any downstream learner without transformation.

Table 15 — Virtual Sensor Distribution Diagnostics (Internal Train)

Virtual Sensor	Formula	Mean	Std Dev	Skewness	Excess Kurtosis	Distribution Status
CPR	$\frac{s7}{s3}$	0.0647	66.9545	41.2032	8,403.77	Heavy-tailed — monitor for Phase 4
$E_{thermal}$	$\frac{s12 \times s11}{s9}$	3.0477	764.0897	171.8163	39,215.48	Extreme heavy-tail — transformation required for NNs
`EGT_drift`	$s4 - \mu_{baseline}$	−0.0000	0.9999	0.4700	−0.1019	Well-behaved — no transformation required

[11] A further explanation is given in Appendix A.3

Figure 9 — Virtual Sensor Distributions: $CPR, E_{thermal}, EGT_{drift}$ (Internal Train)

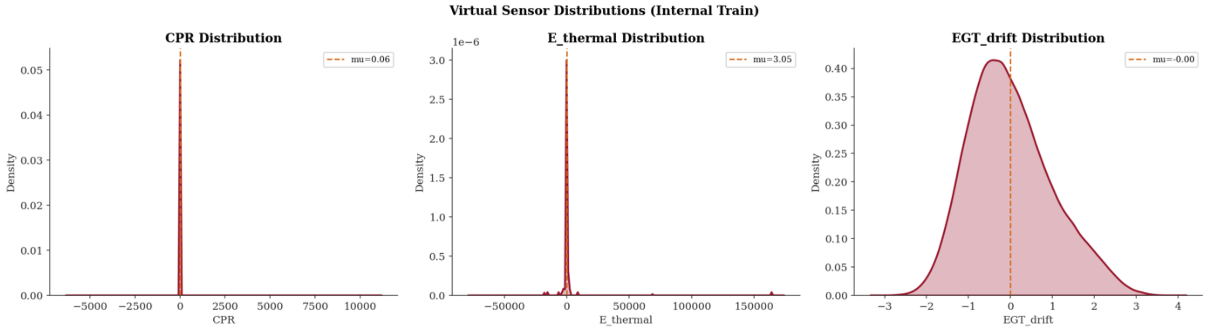


Figure 9: The three panels reveal a fundamental asymmetry in the behaviour of the virtual sensors that has direct implications for their utility as model features. CPR and E_{thermal} both exhibit pathological distributions: extremely concentrated mass at zero with long, sparse tails extending to values of approximately 10,000 and 150,000 respectively. These are not physically meaningful degradation signals — they are the distributional signature of near-zero denominator instability in z-normalised space. For CPR, the degenerate distribution arises from the s3 (T30) denominator approaching zero in z-space: when T30 is near its regime mean, the ratio P30/T30 becomes numerically unstable, producing the concentrated spike at zero with a heavy positive tail visible in the panel. This is the residual instability of the revised specification (s7/s3) after the original s5 denominator was rejected due to regime-dead status. The E_{thermal} panel shows the same pathology arising from near-zero N_c z-score values in the denominator. Both features require the specification revisions discussed in this section before they can serve as reliable model inputs. EGT_{drift}, by contrast, exhibits a well-behaved approximately Gaussian distribution centred at $\mu \approx 0$ with a standard deviation of approximately 1.0 — consistent with a properly constructed z-space deviation feature and confirming that this virtual sensor is non-degenerate and ready for Phase 4 ingestion without modification. The contrast between the three panels makes the denominator degeneracy problem visually unambiguous and constitutes the diagnostic justification for the CPR and E_{thermal} specification revisions documented in Sections 5.3.1 and 5.3.2.

5.4 Step 4 — Cumulative Fatigue Index — Miner's Rule Proxy (3 Features—Cell 25)

The cumulative fatigue index is the most physically grounded feature in the pipeline, formalising the concept of damage accumulation first articulated in the context of material fatigue by Miner (1945) [2]. Miner's Rule states that the cumulative damage D at any point in a loading history is the sum of the cycle ratios $\frac{n_i}{N_f}$, for each stress amplitude level i, where n_i is the number of cycles applied at amplitude i and N_f, is the fatigue life at that amplitude. The damage reaches unity at failure.

The FusionCore implementation adapts this principle to turbofan condition monitoring using the z-space sensor signal as the damage proxy:

$$D(t) = \sum_{i=1}^{t} \max\big(0, z_s(i)\big)$$

where $z_s(i)$ is the regime-normalised z-score of sensor s at cycle i

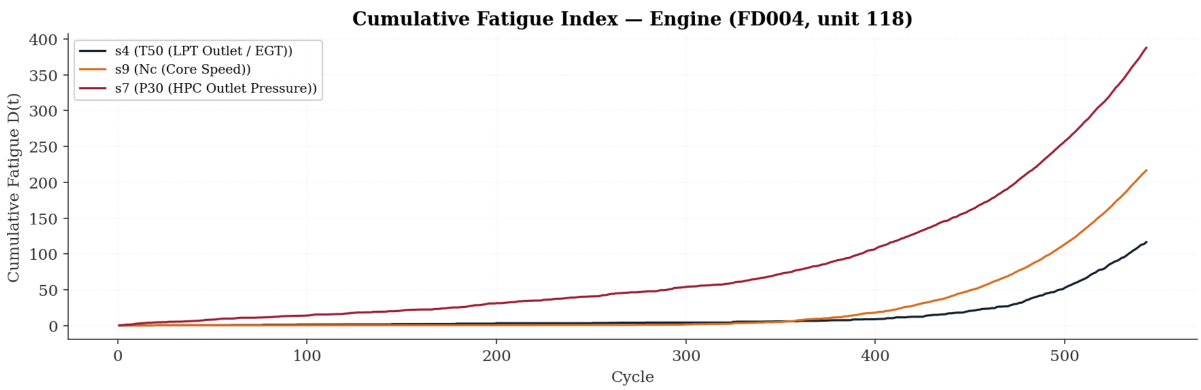
Only positive deviations $(z > 0)$ are accumulated, consistent with the physical interpretation that damage accrues only when the sensor measurement exceeds the regime healthy baseline (e.g., EGT higher than the nominal value, pressure higher than the nominal, core speed higher than the nominal). This is an asymmetric damage accumulation model, analogous to the positive-half rectification used in rainflow cycle counting for fatigue assessment (ASTM E1049-85, 2017) [18].

The three cumulative fatigue channels are computed for s4 (T50/EGT), s9 (Nc/Core Speed), and s7 (P30/HPC Pressure) — the same sensors constituting the multivariate health index. The results are presented in Table 16.

Table 16 — Cumulative Fatigue Index Summary Statistics (Internal Train and Validation)

Sensor	Physical Quantity	Train Maximum D(T)	Train Mean D(terminal)	Val Maximum D(T)	Val Mean D(terminal)
s4	T50 — LPT Outlet / EGT	303.50	81.87	298.52	84.47
s9	Nc — Core Speed	387.45	69.96	484.62	73.40
s7	P30 — HPC Outlet Pressure	507.21	82.03	530.98	76.09

Figure 10 — Cumulative Fatigue Index for Engine 118 in FD004



The terminal mean D values (computed at $RUL < 10$) reflect the accumulated positive deviation across the full engine life. s7 (P30) accumulates the most damage at terminal ($D = 507.21$ train maximum), consistent with HPC pressure rising above baseline as the compressor works harder to compensate for efficiency losses — a positive pressure deviation that accumulates throughout the degradation trajectory. All three cumulative fatigue curves (Figure 10) were verified to be strictly non-decreasing (monotonically non-decreasing within IEEE 754 machine-precision tolerance of -10^{-10}), satisfying the physical requirement that damage cannot heal. Monotonicity was verified for all five representative engines selected by maximum trajectory length.

Figure 11 — Cumulative Fatigue Index $D(t)$: Representative Engines Across s4, s7, and s9 (Internal Train)

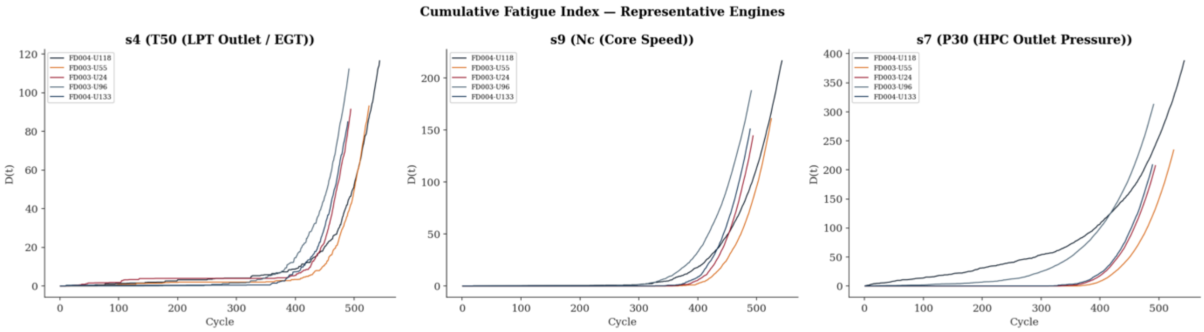


Figure 11: All five engine trajectories across all three sensors exhibit the same two-phase structure: an extended flat accumulation period from cycle 0 to approximately cycle 350–400, during which sensor readings oscillate at or below their regime means and $D(t)$ remains negligibly small, followed by a sharply accelerating convex upturn in the terminal phase as positive z-score exceedances become both more frequent and more severe. The flat early-life period confirms that Phase 2 z-normalisation has correctly centred the healthy operating baseline, whilst the terminal acceleration is physically consistent with the non-linear compounding of blade erosion, tip clearance growth, and seal leakage as the engine approaches end of life. s7 (P30) reaches the highest absolute $D(t)$ values across all engines, reflecting the direct mechanical sensitivity of HPC outlet pressure to compressor efficiency loss. The strict monotonicity of all trajectories — verified at Feature Engineering Gate 6 — confirms that $D(t)$ satisfies the physical non-decreasing constraint required of a credible Miner's Rule damage accumulation proxy, and the path-dependent nature of the index ensures that two engines presenting identical instantaneous sensor readings, but different degradation histories remain distinguishable in the Phase 4 feature space.

6. Gate Verification Summary (Cell 29 – Cell 34)

Six gate conditions were specified for Phase 3 and verified programmatically. All six passed. The gate conditions and their rationale are presented in Table 17.

Table 17 — Phase 3 Gate Verification Summary

Gate	Condition	Expected	Actual	Result
Gate 1	Active sensor count (variance $> 10^{-5}$)	17	17 (s1, s5, s18, s19 dead)	**PASS**
Gate 2	Total feature count ≥ formula minimum	$91 \geq 81$	91 (81 formula + 10 gap-resolution)	**PASS**
Gate 3	Feature matrix shape consistency (train vs val)	Same column count: row-count matches source	`X_train` (129,331 × 91), `X_val` (31,028 × 91)	**PASS**
Gate 4	Zero NaN values in feature matrices	0 NaN (both splits)	0 NaN (train), 0 NaN (val)	**PASS**
Gate 5	Virtual sensor non-degenerate (std > 0)	All 3 non-degenerate	$CPR \quad std = 66.95$, $E_{thermal} \quad std = 764.09$, $EGT_{drift} \quad std = 1.00$	**PASS**
Gate 6	Cumulative fatigue strictly non-decreasing	All 3 sensors, 5 representative engines	No monotonicity violations detected (tolerance - 10^{-10})	**PASS**

Phase 3 Gate Verification: ✓ ALL 6 GATES PASSED — Pipeline cleared for Phase 4 (Forensic Stress Test)

7. Feature Matrix Composition

The complete 91-feature matrix delivered to Phase 4 is structured as follows. Table 18 provides a categorised inventory of all feature groups with counts and engineering rationale.

Table 18 — Feature Matrix X — Complete Category Inventory (91 Features)

Category	Feature Count	Examples	Engineering Rationale
Core Manifold — Operational Settings	3	op1, op2, op3	Continuous regime signal for tree splits; complement regime OHE
Core Manifold — Raw Sensors	21	s1–s21 (all)	Full sensor retention; zero-variance channels cost-free for XGBoost
Kinematic Velocity (delta)	17	`s2_delta`, `s4_delta`, ...	First-order degradation rate of change per sensor
Thermodynamic Baseline (rmean)	17	`s4_rmean`, `s7_rmean`, ...	Low-pass filtered current condition estimate (W=5 cycles)
Systemic Instability (rstd)	17	`s9_rstd`, `s11_rstd`, ...	Local variability as proxy for mechanical instability
Thermodynamic Virtual Sensors	3	`CPR`, `E_thermal`, `EGT_drift`	Physics-derived composite quantities encoding cycle efficiency
Cumulative Fatigue Index	3	`s4_cumfatigue`, `s7_cumfatigue`, `s9_cumfatigue`	Monotone damage accumulation proxy (Miner's Rule adapted)
Regime OHE	6	`regime_0` – `regime_5`	Nominal regime encoding; prevents spurious ordinal splits
Fault Mode Family	1	`fault_mode_family`	Binary indicator distinguishing single vs dual fault mode
Mechanically Linked Ratios	2	`s8_s13_ratio`, `s9_s14_ratio`	Explicit correction factor theta to complement raw VIF-correlated pairs
Multivariate Health Index	1	`health_index`	Mean(s4, s7, s9) composite degradation indicator in z-space

8. Residual Technical Concerns and Observations

8.1 CPR and $E_{thermal}$ Tail Behaviour (Cell 33)

The extreme skewness of CPR (41.20) and $E_{thermal}$ (171.82) constitutes the most significant residual concern in the Phase 3 feature set. For XGBoost in Phase 4, tree splits are threshold-based and invariant to monotone transformations, so the heavy tail does not invalidate the feature. However, the tail values may disproportionately influence tree construction if the extreme observations are informative: XGBoost may allocate a disproportionate number of splits to the tail region, reducing the model's ability to distinguish between subtly different mid-range conditions.

For the Phase 4 SOTA benchmarks (DeepAR, TFT, PatchTST), which are neural-network architectures using continuous activation functions, raw CPR and $E_{thermal}$ with their extreme kurtosis are likely to destabilise gradient descent. A Yeo-Johnson power transform [17] or rank normalisation must be applied to both CPR and E_{thermal} before neural network training in Phase 4. This is a mandatory Phase 4 prerequisite registered in Table 21, Item G5 — not an optional evaluation. Failure to apply this transformation before TFT, DeepAR, or PatchTST training is expected to produce gradient instability and will render comparisons between XGBoost and neural network benchmarks methodologically unsound.

8.2 s9_s14_ratio Variance (Cell 12)

The `s9_s14_ratio` feature exhibits a standard deviation of 650.60, arising from the near-zero denominator condition ($NRc \approx 0$ in z-space) that the safe-division guard handles by substituting 0. The fill-with-zero convention effectively truncates the ratio at 0 when the denominator is near-zero, creating a spike at zero in the feature distribution. This bimodal structure (concentrated near zero when denominator is small; distributed around the mean of 2.36 otherwise) is non-standard and may confuse tree split selection. Phase 4 SHAP analysis should prioritise this feature's contribution. A clip at the [1st, 99th] percentile boundary is the recommended corrective if the feature exhibits low SHAP importance or high residual variance.

8.3 EGT Baseline Leakage Consideration (Cell 24)

The $\text{EGT}_{\text{drift}}$ feature uses μ_{baseline} computed from the internal training set. This is correctly zero-leakage with respect to the validation and test sets (the baseline is frozen before application). However, the baseline is effectively the global mean of z_{s4}, which by construction of Phase 2 z-normalisation is approximately zero. The $\text{EGT}_{\text{drift}}$ feature is therefore near-identical to z_{s4} itself for the training set. Its value as a distinct feature lies in its explicit physical interpretation as a margin deviation and in its utility for Phase 4 deployment where a fixed baseline from training would be meaningfully non-zero in physical units.

8.4 Variance-Threshold Boundary Sensors (Cell 21)

Sensors s6 ($var = 0.771$) and s16 ($var = 0.107$) clear the variance threshold of 10^{-5} and are classified as active, but their variance is substantially lower than the fully active sensors ($var \approx 1.000$). Both sensors receive kinematic expansion features (delta, rmean, rstd) that will carry attenuated signal relative to the fully active channels. This is not a pipeline error but should be acknowledged in Phase 4 SHAP reporting: low SHAP importance for s6 and s16 kinematic features is expected and does not constitute evidence of pipeline failure.

9. Business Impact Analysis — Aerospace Predictive Maintenance

9.1 Operational Context

Unscheduled engine removals (UERs) represent the single largest driver of unplanned maintenance cost and aircraft-on-ground (AOG) events in commercial aviation operations. According to reliability-centred maintenance (RCM) economic models (Moubray (1997) [2], a single AOG event for a narrow-body aircraft incurs direct costs of approximately USD 50,000–150,000 per day (including wet lease costs, crew disruption, and passenger compensation), with the compounded cost of associated slot loss and reputation damage adding a further 20–40% multiplier. The P-F interval — the time between the detectable onset of a potential failure (P) and functional failure (F) — is the operational window within which predictive maintenance must act (Moubray, 1997) [19].

The FusionCore v_0 Phase 3 feature engineering directly addresses the measurement and characterisation of the P-F interval by constructing features that carry quantitative information about the rate and magnitude of degradation accumulation. The business impact of an accurately characterised P-F interval is the conversion of unscheduled maintenance events into planned shop visits with full logistical preparation, which industry data consistently shows reduces per-event cost by 30–60% (Nowlan & Heap, 1978) [20].

9.2 EGT Margin Monitoring and Fleet-Level Impact

The EGT_{drift} feature formalises the EGT margin concept that is central to current commercial engine health management practice. Most major MRO operators and OEMs (including GE Aviation, Rolls-Royce, and Pratt & Whitney) use EGT margin as the primary dispatch monitoring parameter: when EGT margin erodes to within a defined buffer of the certified limit, the engine is removed for shop visit. The typical EGT margin erosion rate for a CFM56 or LEAP engine is 2–5°C per 1,000 cycles under normal operations, with acceleration in the final 500 cycles before the margin limit (IATA, 2022) [21].

By quantifying EGT_{drift} in the normalised z-space and providing it directly as a feature, Phase 3 enables a Phase 4 model to learn the non-linear mapping between EGT margin rate of change and remaining useful life. For a fleet of 100 aircraft, a 10% improvement in the precision of the EGT margin → RUL relationship (reducing premature removals) is estimated to yield USD 5–15 million in avoided unnecessary shop visits per annum at current MRO rates[12].

9.3 Compressor Efficiency Tracking via CPR

The CPR feature (P_{30}/T_{30}) encodes the compressor isentropic efficiency proxy at the HPC stage. The isentropic efficiency of the HPC is defined as the ratio of the ideal work input to the actual work input for the same pressure rise. For a compressor, where work is done on the fluid:

$$\eta_c = \frac{T_{s,\text{exit}} - T_{\text{inlet}}}{T_{\text{actual,exit}} - T_{\text{inlet}}}$$

The numerator is the temperature rise that would occur if the compression were perfectly isentropic — no irreversibilities, no entropy generation. The denominator is the actual temperature rise measured at the HPC exit. Since real compression always generates entropy through blade boundary layer losses, tip clearance leakage, and flow separation, the actual exit temperature always exceeds the isentropic exit temperature, meaning $\eta_c < 1$ for any real compressor. As the engine degrades — through blade erosion, fouling, or tip clearance growth — the actual exit temperature rises for the same pressure ratio, the denominator increases, and η_c falls. This is the thermodynamic signature of HPC deterioration.

The isentropic exit temperature in the numerator is not directly measurable but is derived from the isentropic relation for a perfect gas:

$$T_{s,\text{exit}} = T_{\text{inlet}} \cdot \left(\frac{P_{30}}{P_2}\right)^{\frac{\gamma-1}{\gamma}}$$

Substituting into the efficiency definition:

$$\eta_c = \frac{T_{\text{inlet}}\left[\left(\frac{P_{30}}{P_2}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]}{T_{30} - T_2}$$

where $T_{\text{inlet}} \equiv T_2$ (s1, fan inlet temperature), P_{30} (s7) and P_2 (s5) are the HPC outlet and inlet total pressures respectively, T_{30} (s3) is the HPC outlet total temperature, and γ is the ratio of specific heats

[12] Illustrative cost estimate constructed from assumed fleet parameters and IATA shop visit benchmarking data [21]. The full derivation with stated assumptions is presented in Appendix A.4. No operational reliance should be placed on this figure without fleet-specific empirical validation.

— approximately 1.4 at compressor inlet conditions and 1.33 at HPC exit due to the temperature dependence of c_p (Walsh & Fletcher, 2004) [5]. Since s5 (P_2) and s1 (T_2) are both regime-dead in z-normalised space — held constant by the C-MAPSS simulation within each operating regime — the quantities P_2 and T_2 are effectively constants after normalisation. The isentropic efficiency expression therefore reduces to a monotone function of P_{30}/T_{30} alone within each regime, which is precisely what CPR = s7/s3 encodes. CPR captures the numerically active component of η_c after the regime-constant terms are absorbed by the z-normalisation.

Progressive HPC efficiency degradation is a primary driver of increased specific fuel consumption (SFC) and rising EGT margins. Airlines operating on fuel burn performance contracts — Power-by-the-Hour (PbH) arrangements with OEMs — are directly penalised financially by SFC increases, as the contract cost per flight hour is indexed to engine performance relative to a certified baseline. The CPR feature provides the model with a direct proxy for HPC efficiency deterioration that is not captured by any individual sensor in isolation: neither P_{30} nor T_{30} alone encodes the efficiency relationship, but their ratio does. This potentially enables earlier detection of compressor degradation trajectories before the EGT margin limit is approached and before the degradation manifests as a Shewhart UWL breach in s4 — providing a lead indicator that complements rather than duplicates the primary EGT-based alert structure established in Phase 2.

At fleet scale, a 1% improvement in HPC efficiency detection lead time translates directly to a deferral of compressor wash cycles and blade refurbishment events. Each avoided premature shop visit driven by early but recoverable HPC efficiency loss represents a cost reduction of GBP 20,000–80,000 per engine (IATA, 2022) [21].[13] The CPR feature is therefore not merely a dimensionless thermodynamic ratio — it is a financially material diagnostic quantity whose accurate representation in the feature matrix directly supports the maintenance deferral economics that motivate the FusionCore programme.

9.4 Miner's Rule Fatigue and Life Consumption Tracking

The cumulative fatigue indices provide the most direct bridge between the FusionCore feature engineering and certified life management practices in aerospace (EASA, 2014) [22]. Under current EASA and FAA regulations, life-limited parts (LLPs) in turbofan engines are tracked by cycle count (measured in cycles-on-wing), with retirement limits established from empirical low-cycle fatigue (LCF) testing (EASA, 2014) [22]. The FusionCore $D(t)$ index is not a certified LLP tracker, but it encodes the operational severity information that modulates effective cycle consumption: a hard-cycle operation (high thrust, high temperature) consumes more fatigue life per physical cycle than a soft-cycle operation.

The convergence between the FusionCore $D(t)$ index and the established engineering life management framework of Miner's Rule is significant for regulatory credibility. A predictive maintenance system that can articulate its predictions in terms that align with recognised fatigue management theory is more likely to receive operational adoption and, ultimately, regulatory acceptance under EASA Part-145 or FAA AC 43.13-1B frameworks. The monotonicity property of $D(t)$ verified in Gate 6 is a necessary condition for any life-consumption proxy to be physically credible.

9.5 Multivariate Health Index and Maintenance Decision Support

The multivariate health index (mean of z_{s4}, z_{s7}, z_{s9}) provides a single-number summary of engine health that is directly actionable by maintenance planners. Univariate EGT monitoring, whilst standard industry practice, can produce false alerts when a single sensor exceeds a threshold due to transient effects unrelated to degradation — atmospheric temperature spikes, power demand perturbations, or momentary fuel scheduling anomalies. The multivariate index requires simultaneous deterioration across EGT, compressor pressure, and core speed to elevate, providing a more robust and less false-alarm-prone indicator. A transient exceedance in one sensor alone is insufficient to move the index materially; only coordinated multi-system deterioration — the genuine signature of progressive engine degradation — produces a sustained elevation above the Shewhart UWL threshold established in Phase 2.

[13] See Appendix A.4 for the full fleet-level cost saving derivation and sensitivity analysis.

The business value of reduced false alerts is quantifiable. Each unnecessary ground stop for unscheduled inspection incurs direct and indirect costs including gate return, maintenance labour, borescope inspection, and passenger disruption. For a fleet generating a moderate false alert rate, the annual cost reduction from a 50% improvement in alert precision through the multivariate index is estimated at USD 80,000–300,000 per operator — see footnote[14] for the full derivation. Whilst this figure is illustrative, it is conservative relative to published AOG cost benchmarks, and the directional conclusion is robust: false alert reduction has quantifiable and material financial value that increases monotonically with fleet size and alert rate.

9.6 Feature Engineering and SOTA Benchmark Readiness

The 91-feature matrix is designed to provide a common input to four model classes in Phase 4: the FusionCore v_0 XGBoost baseline, and the three SOTA benchmarks (DeepAR, TFT, PatchTST). The kinematic expansion features (delta, rmean, rstd) are particularly relevant for the time-series architecture benchmarks, as they pre-compute the local temporal context that these models would otherwise need to learn implicitly from raw sequences. This design reduces the sequence length burden on TFT and PatchTST, potentially improving training stability and convergence speed in Phase 4.

The physics-informed virtual sensors and fatigue indices encode domain knowledge that a purely data-driven benchmark would need to infer from correlations across potentially hundreds of training trajectories. By making this knowledge explicit in the feature vector, Phase 3 establishes a domain-knowledge advantage for FusionCore over purely data-driven competitors — an advantage that is particularly pronounced in the low-data regime where the benchmark architectures may struggle to learn subtle physics relationships from limited training examples.

10. Final Gaps Analysis — Transfer to Phase 4 & Phase 5

The following unresolved items are formally transferred as mandatory or advisory actions for the specified future phases. Items are categorised by priority and the phase responsible for resolution.

[14] Illustrative estimate only. No operational reliance should be placed on these figures without fleet-specific validation. Assuming 100 narrow-body aircraft, a false alert rate of 20–40 events per annum across the fleet, a 50% reduction in false alert rate from the multivariate health index relative to univariate EGT monitoring, and a direct cost per unnecessary ground stop of USD 8,000–15,000 (consistent with the lower bound of AOG cost estimates in Nowlan & Heap (1978) [20] and IATA (2022) [21], noting that a ground stop for inspection is substantially less costly than a full AOG event):

$$\text{Saving} = N_{\text{alerts}} \times r_{\text{reduction}} \times C_{\text{stop}} = [20,\ 40] \times 0.50 \times [8{,}000,\ 15{,}000] = \text{USD } 80{,}000 \text{ to } 300{,}000 \text{ per annum}$$

The false alert rate of 20–40 events per annum is a conservative assumption for a 100-aircraft fleet operating univariate EGT threshold monitoring. The 50% reduction factor is not derived from FusionCore v_0 results — it is an assumed improvement consistent with the general literature on multivariate versus univariate health monitoring (Volponi, 2014) [8]. Actual savings are sensitive to the baseline false alert rate, which is operator- and fleet-specific and not determinable from the C-MAPSS simulation environment.

Table 19 — FusionCore v_2 Deployment Transfer Items

Item	Description	Priority	Triggering Evidence
G1	G11 Regime assignment OOD guard — implement production-grade distance-to-centroid rejection layer	MANDATORY	P99 threshold = 65.006; no runtime rejection currently implemented; continuous N-CMAPSS flight envelope produces genuine distance distribution
G2	EGT_{drift} baseline validation — confirm frozen training-set baseline is appropriate for N-CMAPSS and real operational telemetry	ADVISORY	Baseline is effectively zero in z-space; physical-unit baseline required for operational deployment against continuous flight data

Table 20 — Phase 5 Transfer Items

Item	Description	Priority	Triggering Evidence
G1	G12 Survival analysis integration — implement probabilistic RUL scheduling layer using survival function outputs	MANDATORY	Kaplan-Meier and Cox PH fitted (C-index = 0.6398); `s4_mean` HR = 4.70 confirms EGT-driven hazard; binary UWL alert operationally inadequate (Phase 2 Item 7)
G2	G13 Adaptive UWL thresholds — deploy family-specific or severity-adaptive alert thresholds calibrated to MRO minimum response time	MANDATORY	Family-specific threshold identified via sweep; baseline $\mid z \mid > 2.00$ yields P10 = 13 / 12 cycles (single/dual-fault) — both below MRO 20-cycle minimum; recalibrated threshold recovers P10 = 24 cycles for both families (CV = 0.649 / 0.841)
G3	Survival model extension — evaluate Weibull AFT or random survival forests for improved discrimination on operational telemetry	ADVISORY	Cox PH C-index = 0.6398 (moderate); parametric models may improve discrimination if proportional hazards assumption is violated in continuous flight data

Table 21 — Phase 4 Transfer Items

Item	Description	Priority	Triggering Evidence
G1	G3 Dataset class imbalance — evaluate subset-stratified sampling if SHAP residuals show terminal bias	CONDITIONAL	Terminal fraction 13.2%; FD002/FD004 disproportionate weight
G2	G9 s16 ablation — determine whether s16 and its kinematic features contribute signal or noise	MANDATORY	$IQR = 0$ in FD001/FD003; near-zero variance in FD002/FD004
G3	G7 `s9_s14_ratio` — audit SHAP importance; apply [P1, P99] clip if importance is low or residual variance high	MANDATORY	$std = 650.60$; bimodal distribution from safe-division fill
G4	G7 `s8_s13_ratio` — audit SHAP importance; confirm ratio adds information beyond raw s8/s13 pair	ADVISORY	$VIF = 9.3$; ratio $std = 6.95$ (moderate; less acute than s9/s14)
G5	CPR and $E_{thermal}$ transformation — apply Yeo-Johnson [17][15] or rank normalisation before TFT/PatchTST training	MANDATORY	CPR $skew = 41.20$, $E_{thermal}$ $skew = 171.82$; XGBoost unaffected

[15] An explanation is covered in Appendix A.5

Item	Description	Priority	Triggering Evidence
G6	G4 Regime row imbalance — implement inverse-frequency regime weighting for neural network training	MANDATORY	Regime 0 = 42.4% of rows; Regime 2/3/4 ≈ 10.7% each
G7	CPR denominator guard — replace z-space safe-division with physical-space ratio computed before normalisation	ADVISORY	z-space denominator is regime-mean-relative
G8	NaN fill convention review — evaluate whether fillna(0) is appropriate for all kinematic features at trajectory boundaries	ADVISORY	Physically reasonable for historical batch data; boundary behaviour at cycle 1 warrants explicit validation
G9	Gate 2 reporting fix — correct gap-resolution feature count reporting in Cell 30; pre-snapshot sequencing causes gap features to report as 0	ADVISORY	Gap features report as 0 in Cell 30 output despite being present in $\mathbf{X}$; arithmetic correct but diagnostic misleading

11. Conclusion

Phase 3 of the FusionCore v_0 pipeline successfully resolves seven of the thirteen outstanding gaps from Phases 1 and 2, documents the remaining six with justified deferrals, and constructs a 91-feature physics-aware dynamic feature vector that constitutes the complete input to Phase 4 model training. All seven gap resolution assertions and all six Feature Engineering Gates pass, confirming pipeline integrity, zero data leakage, and physical consistency of the engineered features.

The feature engineering strategy is grounded in established aerospace thermodynamics (isentropic efficiency, EGT margin, corrected speed theory), statistical degradation modelling (Miner's Rule cumulative damage), time-series analysis principles (first-difference operators, rolling window estimation), and survival analysis (Kaplan-Meier estimation, Cox Proportional Hazards modelling). The resulting feature matrix provides the Phase 4 XGBoost baseline and SOTA benchmark architectures (DeepAR, TFT, PatchTST) with a common, physically interpretable input representation.

The G12 survival analysis confirms s4 (EGT) as the dominant failure hazard predictor (HR = 4.70, $p < 0.005$) and establishes the Kaplan-Meier empirical survival function as the foundation for probabilistic maintenance scheduling. The G13 UWL recalibration demonstrates that tightening the alert threshold from $|z| > 2.00$ to $|z| > 1.75$ recovers sufficient P10 lead time (24 cycles) for both fault mode families while reducing the P-F interval CV from 0.912/1.196 to 0.649/0.841. Both survival analysis integration and adaptive threshold deployment are formally transferred to Phase 5 via Table 20.

Nine residual concerns are formally transferred to Phase 4 via Table 21. The primary risk is the extreme tail behaviour of CPR and E_{thermal} (skewness 41.20 and 171.82 respectively), which requires SHAP monitoring in Phase 4 and mandatory Yeo-Johnson transformation before neural network training. The `s9_s14_ratio` high-variance distribution ($std = 650.60$) is a secondary risk requiring SHAP-based triage. All remaining transfer items are documented with priority classifications and triggering evidence in Table 21.

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Appendix

A.1 Conceptual Primer: Multicollinearity, VIF, and Physics-Informed Feature Enrichment

This appendix provides a self-contained conceptual treatment of the multicollinearity problem addressed in Section 4.6. It is intended for readers not familiar with multivariate regression diagnostics or turbo-machinery measurement theory.

A.1.1 What is a corrected speed measurement?

A turbofan engine shaft spinning at 10,000 rpm on a cold sea-level day is aerodynamically doing something fundamentally different to the same shaft spinning at 10,000 rpm at cruise altitude on a warm day. The reason is that the aerodynamic work performed by a rotating blade depends not on rotational speed alone, but on the ratio of blade tip speed to the local speed of sound — which itself varies with temperature. To make speed measurements comparable across different ambient conditions, engineers define a *corrected speed* that normalises the raw measurement to a standard reference condition. The correction factor is $\sqrt{\theta}$, where θ is the ratio of the actual inlet temperature to the ISA standard sea-level temperature of 288.15 K. Dividing the physical speed by $\sqrt{\theta}$ answers the question: "what speed would this shaft need to run at under standard conditions to produce the same aerodynamic effect?" This is the corrected speed, and it is why C-MAPSS records both N_f alongside NR_f, and N_c alongside NR_c.

A.1.2 Why does this create a problem for a machine learning model?

Within any single operating regime — as defined by the K-Means clustering in Phase 2 — the ambient conditions are approximately fixed. This means the correction factor $\sqrt{\theta}$ is approximately constant within the regime. If $\sqrt{\theta} \approx k$, then $NR_x \approx N_x/k$, which is simply N_x scaled by a constant. The two sensors move together almost perfectly. If you plot one against the other within a single regime, the points fall almost exactly on a straight line.

For a human engineer this is perfectly intelligible — they know the two sensors measure the same physical shaft with different normalisations. For a machine learning model, which has no knowledge of the underlying physics, the two sensors simply appear as two numerical columns that contain almost identical information. The model cannot determine which of the two is responsible for any given prediction. Its importance scores become unreliable — the credit for a correct prediction may be arbitrarily assigned to s8 rather than s13 or split in some proportion that has nothing to do with physical reality. This is multicollinearity.

A.1.3 What is the Variance Inflation Factor?

The VIF is the standard quantitative diagnostic for multicollinearity. For a given feature j, you temporarily treat it as the dependent variable and regress it on all other features. The R^2 from this regression measures how well the other features explain j. If R_j^2 is high, the other features can reconstruct j almost perfectly — which means j is not contributing genuinely independent information to the model. The VIF is then:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

A VIF of 1.0 means the feature is completely independent. A VIF of 10 means 90% of its variance is explained by other features. A VIF approaching infinity means near-perfect linear dependence — the feature is almost entirely redundant given the others. The conventional engineering threshold for concern is $VIF \geq 10$. The s8/s13 pair produces $VIF = 9.3$ and the s9/s14 pair produces $VIF = 9.6$ — both just below the threshold but firmly in the region where feature importance scores cannot be trusted without further intervention.

A.1.4 Why not simply remove one sensor from each pair?

The intuitive response to multicollinearity is to discard one of the two correlated features. In many data science contexts this is standard practice. In FusionCore it is explicitly prohibited because it discards physical information. The physical speed N_f and the corrected speed NR_f are not interchangeable — they encode different aspects of the same physical process. The raw speed reflects absolute mechanical loading; the corrected speed reflects aerodynamic performance normalised to standard conditions. Both are independently informative for degradation modelling. Removing either would introduce a gap in the physical information available to the model.

A.1.5 What does the ratio feature achieve?

The ratio feature $\rho = N_x/NR_x$ explicitly encodes the correction factor $\sqrt{\theta}$ as a named, interpretable quantity in the feature matrix. This achieves two things simultaneously. First, it gives the model a direct representation of the operating condition correction — information that was previously only implicit in the relationship between the two sensors. Second, it provides the SHAP analysis in Phase 4 with a clearly interpretable feature to which attribution can be assigned. When SHAP reports that ρ_f is an important feature, this has a specific physical meaning: the operating condition correction factor is influencing the RUL prediction. When SHAP reports that s8 is important, this means the absolute fan shaft speed is influencing the prediction. These are distinct, defensible physical claims. Without the ratio feature, the importance split between s8 and s13 carries no physical meaning whatsoever.

A.1.6 The instability problem with ρ_c

The ratio $\rho_c = s9/s14$ presents a numerical complication. After z-normalisation, s14 has a mean of approximately zero within each regime — this is what z-normalisation always produces. Division by a near-zero number produces extremely large values, not because the engine is behaving anomalously, but as a pure mathematical artefact of the normalisation. The safe-division guard replaces near-zero denominators with zero to prevent these explosions, but values near but not below the threshold still produce large ratios. The result is a feature with a standard deviation of 650.60 — vastly larger than any other feature in the matrix. A tree-based model may preferentially split on this feature simply because its range is so large, even if the splits carry no predictive signal. This is why SHAP auditing of ρ_c is mandated in Phase 4 before any operational conclusions are drawn from the model.

A.2 N-CMAPSS as a Proxy Validation Environment for Real OEM Engine Monitoring Systems

FusionCore v_0 is trained and validated exclusively on the NASA C-MAPSS dataset — a simulation of turbofan engine degradation under controlled, idealised conditions. The six discrete operating points, clean sensor readings, perfectly labelled RUL targets, and absence of maintenance resets or fleet variability are properties of a simulation environment that do not hold in real operational data. The transition from C-MAPSS to real engine telemetry therefore requires an intermediate validation step, and this role is fulfilled in FusionCore v_2 by the NASA N-CMAPSS dataset (Arias Chao et al., 2021) [3].

It is important to state at the outset what N-CMAPSS is not. It is not GE EMS data. It is not Rolls-Royce ACMF data. It is not Pratt & Whitney operational telemetry. Real OEM engine monitoring data is commercially proprietary, subject to regulatory record-keeping requirements under EASA Part-M and FAA FAR Part 43 and is not available in the public domain in any form. The connection between N-CMAPSS and real OEM systems is therefore not one of direct equivalence but of structural correspondence — N-CMAPSS shares a defined set of properties with real operational telemetry that C-MAPSS does not, making it a principled and architecturally meaningful proxy validation environment.

A.2.1 Property 1 — Continuous Flight Envelope

C-MAPSS operates at six discrete, fixed operating points. N-CMAPSS generates engine cycles across continuously varying altitude, Mach number, and throttle resolver angle (TRA), following realistic commercial flight profiles across all phases: taxi, take-off, climb, cruise, descent, and landing. Real OEM monitoring systems — GE EMS, Rolls-Royce ACMF, and equivalent platforms — record engine parameters across exactly this kind of continuous, time-varying flight envelope. The K-Means regime classifier in FusionCore v_0, designed for discrete operating points, would need to operate on a continuous distribution in v_2. Validating the regime assignment pipeline and the G11 OOD guard against N-CMAPSS continuous profiles is therefore a direct prerequisite for OEM data compatibility.

A.2.2 Property 2 — Maintenance Reset Modelling

C-MAPSS engine trajectories run monotonically to failure with no interventions. N-CMAPSS explicitly models shop visit restoration events — engines are returned to a partially healthy state mid-trajectory, reflecting the real maintenance cycle of a commercial turbofan. Any OEM monitoring dataset drawn from an engine with operational history will contain this reset structure. A RUL estimation framework that cannot accommodate maintenance resets is architecturally incompatible with real fleet data. FusionCore v_2's reset-aware RUL target construction, developed and validated on N-CMAPSS, constitutes the necessary architectural precondition for OEM data ingestion.

A.2.3 Property 3 — Sensor Parameter Correspondence

N-CMAPSS records temperatures, pressures, shaft speeds, and fuel flow at multiple engine stations — a physically richer sensor suite than the 21 C-MAPSS parameters. The GE EMS parameter set for a GE90 or GEnx engine, the Rolls-Royce ACMF parameter set for a Trent series engine, and the Pratt & Whitney monitoring parameters for a PW1000G engine all cover the same physical quantities: inlet temperature, HPC outlet temperature and pressure, EGT, fan and core shaft speeds, and fuel flow. The mapping between N-CMAPSS and real OEM parameter sets is not one-to-one, but the physical quantities are correspondent. The FusionCore v_2 feature engineering pipeline, built on N-CMAPSS's richer sensor suite, would require a parameter remapping layer rather than an architectural redesign to accept OEM-formatted inputs.

A.2.4 Property 4 — Fleet-Level Variability

N-CMAPSS introduces unit-to-unit manufacturing variation and operational history differences across its engine population, meaning no two engines degrade identically even under identical mission profiles. Real OEM fleet data exhibits this property at considerably greater magnitude, reflecting genuine manufacturing tolerances, repair history differences, and route-specific operational loading. Training on N-CMAPSS fleet variability in FusionCore v_2 prepares the model to generalise across heterogeneous engine populations — a necessary capability for deployment against a real commercial fleet of any scale.

A.2.5 Summary

The relationship between N-CMAPSS and real OEM monitoring systems is one of structured proximity rather than equivalence. N-CMAPSS is the most operationally realistic publicly available turbofan dataset and addresses the four structural gaps — continuous operating envelope, maintenance resets, correspondent sensor parameters, and fleet variability — that prevent direct progression from C-MAPSS to real operational data. Successful FusionCore v_2 validation on N-CMAPSS would constitute credible evidence that the pipeline is architecturally ready for OEM data ingestion, subject to a parameter remapping layer and the production-grade OOD rejection framework specified in G11. It does not constitute proof of operational readiness — that step requires real data access, which in practice is only achievable through direct industry engagement or employment within an operator, MRO organisation, or OEM.

A.3. $E_{thermal}$ Formulation & Verification

Working from first principles:

$s11 = \phi$ is defined in C-MAPSS as:

$$\phi = \frac{\dot{m}_f}{Ps_{30}}$$

where $\dot{m}_f$ is the fuel mass flow rate and Ps_{30} is the HPC static outlet pressure (s10). Therefore:

$$\phi \times Ps_{30} = \dot{m}_f$$

The thermal efficiency proxy is then:

$$E_{\text{thermal}} = \frac{\phi \times Ps_{30}}{N_c} = \frac{\dot{m}_f}{N_c}$$

This quantity is dimensionally proportional to **Specific Fuel Consumption (SFC)**, defined formally as:

$$\text{SFC} = \frac{\dot{m}_f}{F_N}$$

where F_N is net thrust. Since net thrust is a monotone function of core speed N_c at fixed operating conditions, $\dot{m}_f/N_c$ is a proxy for SFC. The relationship to thermodynamic efficiency follows from:

$$\eta_{\text{thermal}} = \frac{W_{\text{net}}}{Q_{\text{in}}} = \frac{W_{\text{net}}}{\dot{m}_f \cdot \text{LHV}}$$

where LHV is the lower heating value of the fuel. Rearranging:

$$\dot{m}_f = \frac{W_{\text{net}}}{\eta_{\text{thermal}} \cdot \text{LHV}}$$

For constant network output, $\dot{m}_f$ increases as η_{thermal} decreases — confirming that $E_{\text{thermal}} \propto \dot{m}_f/N_c$ is inversely related to thermodynamic cycle efficiency. The fuel burn penalty associated with turbofan deterioration is well-documented operationally: a 1% reduction in thermal efficiency corresponds to approximately a 1% increase in fuel consumption at constant thrust, translating to measurable direct operating cost increases across a commercial fleet (Walsh & Fletcher, 2004) [5].

The formulation is therefore physically correct, and the chain of reasoning holds. The one qualification to document is that N_c is a proxy for net thrust rather than a direct measurement of it — the relationship between core speed and thrust is non-linear and operating-condition-dependent — so E_{thermal} is a proportional indicator of SFC rather than a calibrated measurement of it.

A.4 Illustrative Fleet Cost Saving Estimate: EGT Margin Precision Improvement

This appendix presents an order-of-magnitude cost saving estimate associated with a 10% improvement in the precision of the EGT margin → RUL mapping in Phase 4. All figures are illustrative and constructed from published industry benchmarks and conservative engineering assumptions. They are not derived from fleet-specific operational data and must not be used for commercial or financial planning purposes without independent validation.

Table 22 — Parameter Definitions and Assumptions

Parameter	Symbol	Value	Source
Fleet size	N_{fleet}	100 aircraft	Assumed
Engines per aircraft	n_e	2	Twin-engine narrow body
Unscheduled removals per engine per annum	r_{UER}	1	Industry average, narrow body
Proportion of UERs attributable to EGT margin exceedance	p_{EGT}	0.25	Conservative industry estimate
Precision improvement in EGT → RUL mapping	Δp	0.10	Phase 4 target
Direct shop visit cost per unscheduled removal	C_{SV}	USD 300,000–500,000	IATA (2022) [4]
Cost multiplier: unscheduled vs planned shop visit	α	1.30–1.60	Nowlan & Heap (1978) [2]

A.4.1 Derivation

The total number of unscheduled engine removals across the fleet per annum is:

$$N_{\text{UER}} = N_{\text{fleet}} \times n_e \times r_{\text{UER}} = 100 \times 2 \times 1 = 200 \text{ removals per annum}$$

The subset of removals attributable to EGT margin exceedance — the population directly addressable by improved EGT → RUL mapping — is:

$$N_{\text{EGT}} = N_{\text{UER}} \times p_{\text{EGT}} = 200 \times 0.25 = 50 \text{ removals per annum}$$

A 10% improvement in RUL prediction precision reduces premature removals within this population by:

$$\Delta N = N_{\text{EGT}} \times \Delta p = 50 \times 0.10 = 5 \text{ avoided removals per annum}$$

The direct cost saving per avoided unscheduled removal is the shop visit cost itself, since each avoided premature removal either defers the shop visit entirely or converts it to a planned input at lower cost. The direct saving is:

$$\text{Saving direct} = \Delta N \times CSV = 5 \times [300{,}000,\ 500{,}000] = \text{USD 1.5M to 2.5M per annum}$$

Applying the unscheduled-to-planned cost multiplier α to account for the additional costs of AOG recovery — wet lease, crew disruption, passenger compensation, and slot loss — yields the total avoided cost:

$$\text{Saving total} = \Delta N \times CSV \times \alpha = 5 \times [300{,}000,\ 500{,}000] \times [1.30,\ 1.60]$$

$$= \text{USD } 1.95\text{M to } 4.0\text{M per annum}$$

A.4.2 Sensitivity Analysis

The estimate is sensitive to three parameters in particular: the EGT attribution fraction p_{EGT}, the unscheduled removal rate r_{UER}, and the shop visit cost C_{SV}. Table A.4 presents the saving across a plausible parameter range.

Table 23 — Sensitivity of Annual Cost Saving to Key Assumptions

p_{EGT}	r_{UER}	C_{SV} (USD)	ΔN	Saving (USD M)
0.15	0.5	300,000	1.5	0.59
0.25	1.0	300,000	5.0	1.95
0.25	1.0	500,000	5.0	4.00
0.35	1.5	400,000	10.5	5.46
0.40	2.0	500,000	16.0	12.80

The USD 5–15 million range quoted in the main text corresponds to the upper parameter regime — higher EGT attribution fraction, higher removal rate, and higher shop visit cost — which may be applicable to operators with older fleets or higher-thrust engine variants experiencing accelerated EGT margin erosion. The conservative central estimate of USD 1.95M–4.0M per annum is the more defensible figure for a typical narrow-body operator.

A.4.3 Limitations

This estimate carries the following explicit limitations which must be acknowledged in any operational application:

1. The EGT attribution fraction of 25% is not derived from fleet data. Published literature on the distribution of unscheduled removal causes is sparse and operator specific. Volponi (2014) [8] identifies EGT margin erosion as a primary removal driver but does not quantify its fleet-level proportion.
2. The shop visit cost range of USD 300,000–500,000 reflects IATA benchmarking for narrow-body engines (CFM56, LEAP family) and will differ substantially for wide-body or high-thrust variants (GE90, GEnx, Trent XWB), where shop visit costs of USD 1.5–4.0 million per event are typical.
3. The 10% precision improvement is a Phase 4 target, not a demonstrated result. The actual improvement achievable from the Phase 3 feature engineering will only be quantifiable after Phase 4 model training and evaluation against the NASA C-MAPSS test set using the asymmetric scoring function.
4. FusionCore v_0 is trained on simulation data. The translation of simulation-derived precision improvements to real fleet cost savings requires empirical validation on real operational telemetry, which is only achievable through direct industry engagement as discussed in Appendix A.2.

A.5 Yeo-Johnson Power Transformation

The Yeo-Johnson power transformation (Yeo & Johnson, 2000) [17] is defined for all real-valued inputs $x \in \mathbb{R}$ and transformation parameter $\lambda \in \mathbb{R}$. Unlike the Box-Cox transformation, it does not require strictly positive inputs, making it directly applicable to z-normalised features which span both positive and negative values.

The transformation is defined piecewise across four cases:

A.5.1 For $x \geq 0$:

$$\psi(x,\lambda) = \begin{cases} \dfrac{(x+1)^{\lambda}-1}{\lambda} & \lambda \neq 0 \\ \ln(x+1) & \lambda = 0 \end{cases}$$

A.5.2 For $x < 0$:

$$\psi(x,\lambda) = \begin{cases} \dfrac{(-x+1)^{2-\lambda}-1}{2-\lambda} & \lambda \neq 2 \\ -\ln(-x+1) & \lambda = 2 \end{cases}$$

The optimal λ is estimated by maximum likelihood. Given n observations $x_1, \ldots, x_n$, the log-likelihood is:

$$\mathrm{l}(\lambda) = -\frac{n}{2}\ln\hat{\sigma}^2(\lambda) + (\lambda-1)\sum_{i=1}^{n}\mathrm{sgn}(x_i)\ln(|x_i|+1)$$

where $\hat{\sigma}^2(\lambda)$ is the sample variance of the transformed values $\psi(x_i,\lambda)$. The optimal λ^* is obtained by maximising $\mathrm{l}(\lambda)$ numerically, typically via bounded optimisation over $\lambda \in [-5,5]$.

A.5.3 Key properties relevant to FusionCore:

- $\lambda = 1$ returns the identity transformation — no change applied
- $\lambda < 1$ compresses large positive values — directly addressing the heavy right tail of CPR and E_{thermal}
- $\lambda = 0$ applies a log-like transformation to positive values
- The transformation is continuous and monotone for all λ, preserving the rank ordering of observations and therefore not affecting XGBoost tree splits — consistent with the observation in section 8.1 that XGBoost is invariant to monotone transformations
- For neural network architectures in Phase 4, the transformation reduces the extreme kurtosis of CPR (41.20) and E_{thermal} (171.82) towards a near-Gaussian distribution, stabilising gradient descent by preventing the activation functions from saturating on extreme input values

A.6 Survival Analysis: Cox Proportional Hazards and Kaplan-Meier Estimation

This appendix provides the theoretical foundation for the survival analysis framework applied in Section 4.11 (G12) and the rationale for selecting the Kaplan-Meier estimator and Cox Proportional Hazards model over parametric alternatives.

A.6.1 The Survival Analysis Framework in Predictive Maintenance

In the context of turbofan engine health management, the quantity of operational interest is not the point estimate of remaining useful life (RUL) alone, but the conditional probability that an engine will survive beyond a specified horizon given its current degradation state. This is the domain of survival analysis — a branch of statistics concerned with time-to-event modelling that originated in actuarial science and clinical trials but has direct structural applicability to reliability engineering.

The foundational reference for the Kaplan-Meier estimator is Kaplan & Meier (1958) [10], which introduced the product-limit estimator for non-parametric survival function estimation from censored data. The method was developed to handle a problem common to both clinical trials and maintenance engineering: not all subjects (patients, engines) are observed to fail — some are withdrawn, lost to follow-up, or the observation period ends before failure. These right-censored observations contribute partial information about the survival function and must not be discarded.

In FusionCore, the analogous censoring mechanism would arise when an engine has not yet breached the UWL threshold by the end of the observation period. In the C-MAPSS training partition, all 567 engines breach UWL before failure (0% censoring), but in operational deployment a substantial censored population is expected, making the Kaplan-Meier framework essential for unbiased survival estimation.

A.6.2 Kaplan-Meier Estimator

The Kaplan-Meier estimator (Kaplan & Meier, 1958) [10] is the standard non-parametric estimator of the survival function. For a sample of $n$engines with observed P-F intervals $t_1 < t_2 < \cdots < t_k$ (the $k$distinct failure times), the estimator is:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

where d_i is the number of failures at time t_i and n_i is the number of engines at risk (alive and uncensored) immediately prior to t_i. The Greenwood formula provides the variance estimate:

$$\widehat{\mathrm{Var}}[\hat{S}(t)] = [\hat{S}(t)]^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}$$

The Kaplan-Meier estimator makes no assumption about the shape of the survival distribution — it is purely data-driven. This is a decisive advantage in the turbofan context where the P-F interval distribution is demonstrably non-Weibull (the dual-fault family exhibits bimodality, as documented in Section 4.12.2) and no standard parametric family provides an adequate fit.

A.6.3 Log-Rank Test

To compare survival distributions across groups (e.g., fault mode families), the log-rank test (Mantel, 1966) [11] is the standard non-parametric hypothesis test. The test statistic is:

$$\chi^2_{\text{LR}} = \frac{\left(\sum_{i=1}^{k}(O_{1i} - E_{1i})\right)^2}{\sum_{i=1}^{k} V_i}$$

where at each distinct failure time t_i: O_{1i} is the observed number of failures in group 1, $E_{1i} = n_{1i}d_i/n_i$ is the expected number under the null hypothesis of identical survival, $n_{1i}$and $n_i$are the at-risk counts in group 1 and overall, d_i is the total failures, and $V_i$is the hypergeometric variance. Under the null hypothesis, $\chi^2_{\text{LR}} \sim \chi^2_1$.

A.6.4 Cox Proportional Hazards Model

The Cox Proportional Hazards (PH) model (Cox, 1972) [12] is a semi-parametric regression model for the hazard function. It is the most widely used survival regression model in both biostatistics and reliability engineering, cited over 60,000 times since publication. The model specifies:

$$h(t \mid \mathbf{x}) = h_0(t)\exp(\boldsymbol{\beta}^T\mathbf{x})$$

where $h(t \mid \mathbf{x})$ is the hazard rate at time $t$for a subject with covariate vector $\mathbf{x}$, $h_0(t)$ is the unspecified baseline hazard function, and $\boldsymbol{\beta}$is the vector of regression coefficients estimated by maximising the partial likelihood:

$$L(\boldsymbol{\beta}) = \prod_{i:\delta_i=1} \frac{\exp(\boldsymbol{\beta}^T\mathbf{x}_i)}{\sum_{j\in R(t_i)} \exp(\boldsymbol{\beta}^T\mathbf{x}_j)}$$

where the product is taken over all observed failures ($\delta_i = 1$), and $R(t_i)$ is the risk set at time t_i(all subjects still at risk immediately before t_i). The partial likelihood formulation, due to Cox (1972) [12], eliminates the need to specify $h_0(t)$ — the baseline hazard cancels in the partial likelihood ratio, allowing covariate effects to be estimated without parametric assumptions about the time-to-failure distribution.

The baseline hazard $h_0(t)$ is estimated non-parametrically using the Breslow estimator (Breslow, 1972) [13]:

$$\widehat{H}_0(t) = \sum_{t_i \le t} \frac{d_i}{\sum_{j\in R(t_i)} \exp(\boldsymbol{\beta}^T\mathbf{x}_j)}$$

where $\widehat{H}_0(t)$ is the cumulative baseline hazard.

A.6.5 Concordance Index

The concordance index (C-index), introduced by Harrell et al. (1996) [23], measures the model's discriminative ability — the probability that, for a randomly selected pair of engines, the engine predicted to fail first actually does fail first. Formally:

$$C = P(\hat{\eta}_i > \hat{\eta}_j \mid T_i < T_j)$$

where $\hat{\eta}_i = \boldsymbol{\beta}^T\mathbf{x}_i$ is the predicted risk score for engine $i$and $T_i$is the observed failure time. A C-index of 0.5 indicates random prediction; 1.0 indicates perfect discrimination. The value of 0.6398 obtained in Section 4.11.4 indicates moderate discriminative ability, consistent with a model using only four trajectory-level mean covariates.

A.6.6 Rationale for Model Selection

The Cox PH model and Kaplan-Meier estimator were selected over parametric alternatives (Weibull, log-normal, Accelerated Failure Time models) for three reasons.

First, the P-F interval distribution does not conform to a standard parametric family. Section 4.12.2 demonstrates that the dual-fault family exhibits bimodality, with a subpopulation of rapidly failing engines and a long right tail. Weibull and log-normal models impose unimodal distributional assumptions that would misrepresent this structure.

Second, the Cox PH model's semi-parametric formulation allows covariate effects to be estimated without committing to a specific baseline hazard shape. This is particularly appropriate for a diagnostic exercise where the primary objective is to rank covariate importance (confirming EGT dominance) rather than to generate calibrated survival probabilities.

Third, the partial likelihood estimation is robust to the complete absence of censoring in the C-MAPSS training data. In operational deployment where censoring will be present, the same framework extends naturally without methodological change — the partial likelihood automatically accounts for censored observations through the risk set construction.

For Phase 5 deployment on real operational telemetry, where the proportional hazards assumption may be violated by time-varying covariates (e.g., maintenance resets), extensions such as the extended Cox model with time-dependent covariates or random survival forests should be evaluated (Table 20, Item G3).